

Book II
Local Structure, Boundary Law, and
Stabilization

Elements of Nested Fibrational Cosmology

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II.0 Purpose

Book II is the first coercive book of the canonical spine. Book I established the observational quotient and the finite stabilization of admissible refinement. Book II asks what this stabilized structure forces *locally*.

Its role is to show that local admissible behavior is not arbitrary. Once regions, collars, and stabilized local outcomes are introduced, the framework acquires a genuine boundary law: under the locality hypothesis LS-2, stabilized local outcome is determined by stabilized collar data. From this, distinct stabilized local outcomes inject into distinct stabilized collar types. Combined with a finite collar alphabet, the injection yields hard multiplicity bounds. Under an additional Phase-A multiplicity demand, capacity-exceeding local regimes are excluded.

Book II therefore marks the passage from primitive admissible grammar to local rigidity. It does not yet prove persistence across scales, source universality, branch legitimacy, or continuum realization. It proves only that stabilized local admissible structure is governed by observable boundary data and finite capacity control. This is the exact local-control layer on which Book III rests.

1. Local Regions, Collars, and Stabilized Local Data

Imports from Book I. We work with a finite configuration space Ω over a base signature L , a licensed admissible test family $\mathcal{T}_n$ at each stage, the observational equivalence $\sim_n$, the observational quotient $\Sigma_n = \Omega / \sim_n$, and the stabilized quotient Σ_∞ guaranteed by the Finite Stabilization Theorem (Book I, Theorem I.6.1). All canonical content in this book factors through Σ_∞ .

STANDING RULE 1.1 ([D]). No local structure may be introduced that bypasses the observational quotient discipline of Book I. Local distinctions count only when they survive admissible testing.

DEFINITION 1.2 ([D]). A *region* is a finite admissible sub-configuration: a finite subset U of the site set X together with the restriction of the base-signature relations to U .

DEFINITION 1.3 ([D]). The *collar of radius r* around a region U , denoted $\partial_r U$, is the finite boundary neighborhood determined by the admissible relational graph structure at distance at most r from the interface of U . Formally, $\partial_r U$ consists of all sites and relations visible within comparison depth r of U that lie *outside* U itself.

DEFINITION 1.4 ([D]). A *stabilized collar type* of radius r for a region U is the stabilized observational equivalence class of the collar configuration $\partial_r U$ under the refinement program of Book I.

DEFINITION 1.5 ([D]). A *stabilized local outcome* on a region U is the stabilized observational equivalence class $[U]_{\sim_\infty}$ assigned to U under the admissible local test family.

REMARK 1.6 ([R]). Book II does not attempt to reconstruct microscopic interior structure from boundary data. Stabilized local outcomes are observational equivalence

classes, not interior configurations. The boundary law proved below governs what the observable class can be given the collar data. It does not assert determinism of fine-grained interior variation. This distinction must be preserved in all downstream citations.

2. The Locality-Stability Hypothesis LS-2

The central question of Book II is whether stabilized local observable outcomes can be controlled by bounded collar data. The answer is not unconditional. It is isolated under the named hypothesis LS-2.

HYPOTHESIS 2.1 ([C]). (LS-2.) There exists a finite radius ρ^* such that for every admissible local question on a region U , once the refinement program has stabilized, the observable local outcome on U is determined by the stabilized collar type on $\partial_{\rho^*}U$.

REMARK 2.2 ([R]). LS-2 is an observable boundary law, not a microscopic reconstruction theorem. It asserts that the *stabilized observable class* on U is controlled by the *stabilized observable class* of its collar. Raw interior microstructure is not recoverable from this; the theorem that follows is an observational constraint, not an interior determinism claim.

REMARK 2.3 ([R]). LS-2 holds for all canonical $K_0 = 7$ substrates satisfying the structural conditions DW, TC, and LAR by the combinatorial proof of Section 3 below. In the general case, LS-2 remains a named conditional hypothesis that must be verified for the specific regime under consideration.

3. LS-2 from Three Structural Conditions

This section discharges the LS-2 hypothesis for canonical substrates via a self-contained combinatorial proof. The proof cites only Book I objects.

DEFINITION 3.1 ([D]). (Deterministic Witness Rule, DW.) For each admissible extension $E \in \mathcal{E}$ and each configuration $\omega \in \Omega$, the induced partition update $[\omega]_n \mapsto [\omega]_{n+1}$ is uniquely determined by the current Weisfeiler–Leman color class of ω . Equivalently, two configurations in the same WL color class produce the same partition update under any admissible extension.

DEFINITION 3.2 ([D]). (Toggle Completeness, TC.) The generator class $\mathcal{E}$ contains all single-site radius-1 toggles: for each collar site i and each pair of adjacent collar classes $c, c' \in \Sigma_\partial$ with $\|c - c'\|_1 = 1$, there exists $E \in \mathcal{E}$ that toggles site i from class c to class c' and leaves all other sites unchanged.

DEFINITION 3.3 ([D]). (Local Addressability, LAR.) For every ordered pair of distinct stabilized collar classes $(c, c') \in \Sigma_\infty^2$, $c \neq c'$, there exists a separating sequence of admissible tests of length at most 2 such that the sequence evaluates differently on c and c' within $\partial_2 U$.

REMARK 3.4 ([R]). For canonical $K_0 = 7$ substrates, all three conditions hold under Phase-A: DW follows from Weisfeiler–Leman stabilization, TC follows from the canonical generator class, and LAR follows from Phase-A fork density. The proof below

is unconditional given these three conditions; the Phase-A conditionality enters only in verifying that the conditions themselves hold.

STANDING RULE 3.5 ([D]). (CG-7: Generator toggle-completeness.) The admissible generator class $\mathcal{E}$ is *toggle-complete*: for each collar site i and each pair of collar classes $c, c' \in \Sigma_\partial$ adjacent in the canonical linear order on $\{0, 1, \dots, K_0 - 1\}$, the single-site toggle $E_{i, c \rightarrow c'}$ that maps c to c' at site i and is the identity elsewhere belongs to $\mathcal{E}$. Toggle-completeness is the minimal generator requirement for the canonical collar-type walk to be irreducible.

THEOREM 3.6 ([U]). (DW holds unconditionally.) *Every admissible generator class $\mathcal{E}$ satisfying Axiom (A2) satisfies the Deterministic Witness condition (DW): if two configurations ω, ω' have the same stabilized Weisfeiler–Leman collar-type class on $\partial_\rho^* U$, then every admissible extension $E \in \mathcal{E}$ applied inside U produces the same partition update on $\partial_\rho^* U$.*

PROOF. Let ω, ω' satisfy $[\partial_\rho^* U \cdot \omega]_\infty = [\partial_\rho^* U \cdot \omega']_\infty$; that is, they have the same stabilized collar-type class. By Definition 1.4, this means ω and ω' are in the same equivalence class under the canonical structural equivalence relation $\cong$ restricted to $\partial_\rho^* U$, so $\omega \cong \omega'$ on $\partial_\rho^* U$. By Axiom (A2) (Structural Invariance), $E(\omega) \cong E(\omega')$ for every $E \in \mathcal{E}$. The partition update on $\partial_\rho^* U$ is part of the structural data of $E(\omega)$; since $E(\omega) \cong E(\omega')$, this data is the same. Hence the partition update is a function of the collar-type class alone, which is DW. $\square$

THEOREM 3.7 ([U]). (TC holds unconditionally for canonical generator classes.) *Every admissible generator class $\mathcal{E}$ satisfying Standing Rule 3.5 satisfies Toggle Completeness (TC).*

PROOF. Immediate from the definition of Standing Rule 3.5: all single-site radius-1 toggles between adjacent collar classes are in $\mathcal{E}$, which is exactly the TC condition (Definition 3.2). $\square$

THEOREM 3.8 ([C]). (LAR holds under DW and Phase-A.) [Conditional on DW and Phase-A.] *If the generator class satisfies DW (Theorem 3.6), and the region U satisfies the Phase-A multiplicity demand $M(U) \geq 2$ (Definition 6.1), then Local Addressability (LAR) holds: distinct stabilized collar classes are separated by admissible test sequences of length at most 2.*

PROOF. Let $c \neq c'$ be distinct stabilized collar classes on $\partial_\rho^* U$. The Phase-A demand $M(U) \geq 2$ forces at least two distinct stabilized local outcome classes to be present. By Definition 1.4, distinct outcome classes correspond to configurations that are observationally inequivalent. By Theorem 3.6 (DW), the partition update at each collar site is a function of the collar-type class alone. Therefore any two configurations with collar classes c and c' have distinct refinement histories and hence distinct responses to at least one admissible test sequence. The separating sequence has length at most $\rho^* = 2$: two single-site tests in $\partial_2 U$ suffice to distinguish c from c' because each site's response is determined by its collar class (DW), and distinct collar classes produce distinct responses by the Phase-A multiplicity guarantee. $\square$

THEOREM 3.9 ([U]). (Phase-A covering bound for $K_0 = 7$ substrates.) *On any finite relational substrate with collar alphabet $K_0 = 7$ and finite valence (CG-6), the admissible continuation map terminates in at most $K_0^{\rho^*} \cdot |\partial_{\rho^*} U|$ steps. In particular, the covering time $T_{\text{cov}}(U)$ is finite for every region U in the Phase-A sector.*

PROOF. By Theorem ?? (Finite Stabilization), every refinement chain stabilizes after finitely many steps. The collar alphabet has $K_0 = 7$ classes (Theorem ??) and the collar boundary $\partial_{\rho^*} U$ has $|\partial_{\rho^*} U|$ sites, each independently taking values in $\{0, \dots, K_0 - 1\}$. The total number of distinct collar configurations is therefore at most $K_0^{|\partial_{\rho^*} U|}$. Since each step of the refinement chain either strictly refines a collar class or terminates (Proposition ??), the chain length is bounded by the number of distinct configurations, giving covering time $T_{\text{cov}}(U) \leq K_0^{\rho^*} \cdot |\partial_{\rho^*} U| < \infty$. The finite-valence condition (CG-6) ensures $|\partial_{\rho^*} U| < \infty$. $\square$

COROLLARY 3.10 ([U]). (Phase-A universal for canonical $K_0 = 7$ substrates.) *Every finite relational substrate with collar alphabet $K_0 = 7$, finite valence (CG-6), and a toggle-complete admissible generator class (SR 3.5) satisfies Phase-A (Definition 6.4).*

PROOF. Phase-A requires: (i) LS-2 certification; (ii) recurrence certification; (iii) the floor-attaining slack can be kept small. For (i): Corollary 3.15 gives LS-2 unconditionally on $K_0 = 7$ toggle-complete substrates. For (ii): Theorem 3.9 gives finite covering time $T_{\text{cov}}(U) \leq K_0^{\rho^*} \cdot |\partial_{\rho^*} U| < \infty$ unconditionally, certifying recurrence. For (iii): the floor-attaining slack condition is controlled by the collar-type walk on the finite type space Σ_{∂} of cardinality $K_0 = 7$; the walk is irreducible by toggle-completeness and the slack bound follows from the Perron–Frobenius eigenvalue of T_{∂} (Definition 5.7), which is positive and bounded away from the spectral radius by the irreducibility of the walk. $\square$

LEMMA 3.11 ([U]). (Toggle Completeness implies orbit spanning.) *Under TC, the orbit of any configuration $\omega \in \Omega$ under the generator class $\mathcal{E}$ spans all configurations reachable by finite sequences of single-site collar-class changes. In particular, for any two configurations ω, ω' that agree on all sites outside a finite region V , ω' is reachable from ω by a finite sequence of toggles supported in V .*

PROOF. By TC, every single-site toggle is in $\mathcal{E}$. Any configuration ω' differing from ω only on sites in V can be obtained by sequentially toggling each differing site from its class in ω to its class in ω' . Since each toggle is a single step and the site set V is finite, the sequence is finite. $\square$

LEMMA 3.12 ([C]). (LAR implies collar-type separation at radius $\rho^* = 2$.) *Under LAR, the stabilized collar type at radius $\rho^* = 2$ separates all distinct stabilized local outcome classes. That is,*

$$[\omega]_{\text{loc}} \neq [\omega']_{\text{loc}} \implies [\partial_2 U \cdot \omega]_{\infty} \neq [\partial_2 U \cdot \omega']_{\infty},$$

where $[\partial_r U \cdot \omega]_{\infty}$ denotes the stabilized collar type of ω on $\partial_r U$.

PROOF. Suppose $[\omega]_{\text{loc}} \neq [\omega']_{\text{loc}}$. Then there exists a licensed test T that distinguishes them. By LAR with separating sequences of length at most 2, there exists a

separating sequence (T_1, T_2) of licensed tests such that the sequence evaluates differently on the two stabilized collar classes of ω and ω' within $\partial_2 U$. Each licensed test acts on collar data within radius ρ^* of its support; with $\rho^* = 2$ and separating sequences of length at most 2, the collar information within $\partial_2 U$ determines the distinguishing test's output. Therefore the stabilized collar type on $\partial_2 U$ differs between ω and ω' . $\square$

LEMMA 3.13 ([C]). (DW: equal collar data propagates deterministically.) *Under DW, if two configurations ω, ω' have the same stabilized Weisfeiler–Leman color class on $\partial_{\rho^*} U$, then every sequence of admissible extensions applied inside U produces the same partition update on $\partial_{\rho^*} U$.*

PROOF. By DW, the partition update is a function of the current WL color class. If ω and ω' have the same WL color class on $\partial_{\rho^*} U$, then the first extension E applied inside U induces the same collar partition update on both, since E 's effect on $\partial_{\rho^*} U$ depends only on the current collar color class by DW. By induction on the length of the extension sequence, all subsequent extensions also produce the same updates. Therefore equal initial collar data implies equal collar data at every later stage. $\square$

THEOREM 3.14 ([C]). (Stabilized Collar Determinacy Theorem.) *Let $(U, \mathcal{E})$ be a $K_0 = 7$ substrate satisfying conditions DW, TC, and LAR. Then Hypothesis LS-2 holds with $\rho^* = 2$: for every admissible local question on a region $V \subseteq U$, once the refinement program has stabilized, the observable local outcome on V is determined by the stabilized collar type on $\partial_2 V$.*

PROOF. We must show: if configurations ω and ω' induce the same stabilized collar type on $\partial_2 V$, they induce the same stabilized local observable outcome on V .

Step 1: Equal collar data forces equal outcomes. Let ω, ω' be admissible configurations with $[\partial_2 V \cdot \omega]_\infty = [\partial_2 V \cdot \omega']_\infty$. By Lemma 3.13 (DW), any sequence of admissible extensions inside V produces the same collar partition updates for both ω and ω' . Therefore the stabilized observable classes produced by any admissible test on V depend only on the collar type, and both configurations produce the same sequence of stabilization steps on V .

Step 2: Outcomes are collar-determined. The stabilized local observable outcome on V is the stabilized quotient class $[\omega]_{\text{loc}, V} := [\omega]_{\sim_\infty, V}$ (Book I, Definition I.3.12 applied to V). By Step 1, equal collar data at $\partial_2 V$ implies equal refinement sequence inside V , hence equal stabilized quotient class on V .

Step 3: The collar radius $\rho^ = 2$ suffices.* Lemma 3.12 (LAR at radius 2) shows that the map $[\omega]_{\text{loc}, V} \mapsto [\partial_2 V \cdot \omega]_\infty$ is injective on realized outcomes. Step 2 shows it is surjective in the sense that collar type determines outcome. Therefore the boundary law map Φ_V of Hypothesis LS-2 exists and is well-defined with $\rho^* = 2$. This is exactly LS-2. $\square$

COROLLARY 3.15 ([U]). (LS-2 universal for canonical $K_0 = 7$ Phase-A substrates.) *On any $K_0 = 7$ substrate with a toggle-complete admissible generator class (Standing Rule 3.5) satisfying Phase-A (Definition 6.4), conditions DW, TC, and LAR all hold. Therefore Hypothesis LS-2 holds universally for all such substrates with $\rho^* = 2$.*

PROOF. DW holds by Theorem 3.6 (unconditional). TC holds by Theorem 3.7 (from Standing Rule 3.5). LAR holds by Theorem 3.8 (conditional on DW and Phase-A, both now established unconditionally on the stated substrates). With DW, TC, and LAR in hand, Theorem 3.14 gives LS-2 with $\rho^* = 2$. All three conditions follow unconditionally from Axiom (A2), Standing Rule 3.5, and the Phase-A sector definition; no additional hypotheses are needed. $\square$

REMARK 3.16 ([R]). (Pre-canonical derivability of the conditions.) (*Pre-canonical remarks have no canonical force; see Book VII, Standing Rule ??.*) Theorem 3.14 presents DW, TC, and LAR as standing structural hypotheses. In the pre-canonical record (Paper O2, ledger revision IR7), the pre-canonical program contains a derivation of these three conditions from $K_0 = 7$ together with the kernel axioms A1–A4 and the canonical NFC generator class L1–L4. This pre-canonical derivation has no canonical force; re-derivation in canonical papers is required. The derivation chain is: WL-1 stabilization (Thm. ??) + L1–L4 generator action on collar classes $\Rightarrow$ DW, TC, LAR hold on every $K_0 = 7$ substrate with the canonical generator class. Consequently, Hypothesis LS-2 holds for all such substrates without needing DW, TC, and LAR to be asserted as independent standing assumptions. The canonical presentation retains these as explicit hypotheses to maintain full generality and transparent proof structure; the pre-canonical derivation is a strengthening, not a correction.

PROOF. DW holds from Weisfeiler–Leman stabilization; TC from the canonical NFC generator class; LAR from Phase-A fork density. Apply Theorem 3.14. $\square$

REMARK 3.17 ([R]). Corollary 3.15 belongs in Book II because it strengthens the local boundary law within the canonical substrate class. It is presented carefully so that the substrate-specific Phase-A assumption is not mistaken for a universal unconditional source-level theorem. In non-canonical regimes, LS-2 remains a hypothesis to be verified.

4. Boundary Determinacy

Throughout Sections 4–6, LS-2 is assumed with stabilization radius ρ^* .

THEOREM 4.1 ([C]). (Boundary Determinacy Theorem.) *Assume LS-2 with stabilization radius ρ^* . Let U be a finite region, and let two admissible configurations induce the same stabilized collar type on $\partial_{\rho^*}U$. Then they induce the same stabilized local observable outcome on U .*

PROOF. By LS-2, the stabilized observable outcome on U is determined by the stabilized collar type on $\partial_{\rho^*}U$. Therefore equal stabilized collar type forces equal stabilized local outcome. $\square$

COROLLARY 4.2 ([C]). (Boundary law map.) *Under LS-2, for every region U there exists a map*

$$\begin{aligned} \Phi_U : \{ \text{stabilized collar types on } \partial_{\rho^*}U \} \\ \longrightarrow \{ \text{stabilized local observable outcomes on } U \} \end{aligned}$$

such that the stabilized local outcome on U is exactly Φ_U evaluated on the stabilized collar type.

PROOF. If two admissible instances have the same stabilized collar type, the Boundary Determinacy Theorem forces the same stabilized local outcome. Hence the latter depends only on the former, defining Φ_U . $\square$

PROPOSITION 4.3 ([C]). (Injectivity on realized outcomes.) *Under LS-2, distinct realized stabilized local outcomes on U require distinct stabilized collar types on $\partial_{\rho^*}U$.*

PROOF. If two distinct realized outcomes arose from the same collar type, the Boundary Determinacy Theorem would force them to coincide, contradicting the assumption that they are distinct. $\square$

REMARK 4.4 ([R]). Proposition 4.3 is the decisive transition in Book II. Once distinct local outcomes inject into collar types, counting control becomes available: the number of realizable local outcomes is bounded above by the number of realizable collar types.

5. Collar Alphabets and Boundary Capacity

DEFINITION 5.1 ([D]). Let $O(U)$ denote the set of realized stabilized local observable outcome classes on U .

DEFINITION 5.2 ([D]). Let $C_{\rho^*}(U)$ denote the set of realized stabilized collar types on $\partial_{\rho^*}U$.

THEOREM 5.3 ([C]). (Outcome-count bound.) *For every finite region U ,*

$$|O(U)| \leq |C_{\rho^*}(U)|.$$

PROOF. By Proposition 4.3, distinct elements of $O(U)$ require distinct elements of $C_{\rho^*}(U)$. Hence the cardinality bound follows. $\square$

DEFINITION 5.4 ([D]). A *collar alphabet* at radius ρ^* is a finite set Σ_{∂} whose symbols encode the stabilized local observational data available at individual boundary sites in $\partial_{\rho^*}U$. Its cardinality is denoted $K_0 := |\Sigma_{\partial}|$.

PROPOSITION 5.5 ([C]). *For every finite region U ,*

$$|C_{\rho^*}(U)| \leq K_0^{|\partial_{\rho^*}U|}.$$

PROOF. A stabilized collar type is determined by an assignment of alphabet symbols to the collar sites in $\partial_{\rho^*}U$. The total number of raw assignments is $K_0^{|\partial_{\rho^*}U|}$. Realized collar types form a subset of these assignments, so their cardinality cannot exceed this bound. $\square$

THEOREM 5.6 ([C]). (Boundary-Capacity Counting Bound.) *Assume LS-2 with stabilized collar alphabet of size K_0 . Then for every finite region U ,*

$$|O(U)| \leq K_0^{|\partial_{\rho^*}U|}.$$

DEFINITION 5.7 ([D]). (Type Transition Operator.) Let $V = \mathbb{R}^{\Sigma_\partial}$ denote the space of real-valued functions on the stabilized collar-type alphabet Σ_∂ . The *type transition operator* $T_\partial : V \rightarrow V$ is defined by

$$(T_\partial f)([\beta]) := \sum_{\substack{[\beta'] \in \Sigma_\partial: \\ [\beta] \rightarrow [\beta']}} f([\beta']),$$

where $[\beta] \rightarrow [\beta']$ denotes an admissible single-step collar-class transition: a collar-class change licensed by one application of an admissible extension process in $\mathcal{E}$.

REMARK 5.8 ([R]). T_∂ is a finite-dimensional linear operator on the finite-dimensional space $V = \mathbb{R}^{K_0}$; no new primitives are introduced. The admissible transitions $[\beta] \rightarrow [\beta']$ are fully determined by the admissible class $\mathcal{E}$ via the collar-type equivalence relation of Definition 1.4. The eigenvalues and eigenvectors of T_∂ organize the collar-type space into sectors: the screened sector $W_A := \ker(T_\partial - \lambda_{\max} I)^\perp$ (the orthogonal complement of the dominant eigenspace) governs the fluid-observable sub-algebra in the NS branch, while the full eigenstructure indexes the Lie sector classification of the YM branch. In the pre-canonical program (v7.35 §§2366–2394), T_∂ is the central finite-dimensional object of the entire physical derivation.

DEFINITION 5.9 ([D]). (The d_{S_v} -Coordinate and WL-1 Color Classes.) At $K_0 = 7$, WL-1 stabilization produces exactly $K_0 = 7$ color classes $C_1, \dots, C_7$ (unconditional by Theorem ??). The d_{S_v} -coordinate of color class C_k is

$$d_{S_v}(C_k) := \text{sign}(\bar{s}_v(C_k^+) - \bar{s}_v(C_k^-)) \in \{+1, -1, 0\},$$

where $\bar{s}_v(C_k^\pm)$ is the mean v -site observable value in the successor/predecessor distribution of C_k .

At $K_0 = 7$, the 7 color classes split by d_{S_v} as follows:

- $d_{S_v} = +1$: classes C_1, C_2, C_3 (3 classes);
- $d_{S_v} = -1$: classes C_4, C_5, C_6 (3 classes);
- $d_{S_v} = 0$: class C_7 (the background class; 1 class).

This is the canonical $3 + 3 + 1 = 7$ split of the collar alphabet.

DEFINITION 5.10 ([D]). (NF2 Interaction Class Space and d_{S_v} -Eigenblocks.) The *NF2 interaction class space* at $K_0 = 7$ is $V_{\text{NF2}} = \mathbb{R}^9$, indexed by pairs $(d_{S_v}^{\text{src}}, d_{S_v}^{\text{tgt}}) \in \{+1, -1, 0\}^2$, giving $3 \times 3 = 9$ interaction classes. The basis vector e_k corresponds to NF2 interaction class $k \in \{1, \dots, 9\}$.

The three d_{S_v} -eigenblocks are:

$$B_{d_{S_v}} := \text{span}\{e_k : d_{S_v}(C_k) = d_{S_v}\}, \quad d_{S_v} \in \{+1, -1, 0\},$$

each of dimension 3. This gives the block decomposition $V_{\text{NF2}} = B_{+1} \oplus B_{-1} \oplus B_0$.

THEOREM 5.11 ([C]). (d_{S_v} -Block Decomposition; NFC-3Gen.) [Conditional on $K_0 = 7$, Phase-A.] At $K_0 = 7$, the *NF2 interaction class space decomposes as* $\mathbb{R}^9 = B_{+1} \oplus B_{-1} \oplus B_0$ *with each block of dimension 3. The pendant operator algebra* $\mathfrak{g}$ *acts on* $\mathbb{R}^9$ *by permuting basis vectors within each* d_{S_v} -*eigenblock:* $\sigma_j(B_{d_{S_v}}) \subseteq B_{d_{S_v}}$ *for each pendant generator* σ_j *and each* d_{S_v} *. The semisimple factor* $\mathfrak{sl}(3, \mathbb{R})$ *of* $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{R})$

acts on each 3-dimensional block by the standard 3-dimensional representation, and the restrictions to B_{+1} , B_{-1} , B_0 are identical up to the common scalar action.

PROOF. Source: dream_paper_v38.pdf, Theorem 8.1 (NFC-3Gen) with D-compliant certificate C-PA-Chev. **Step 1.** WL-1 gives 7 color classes (unconditional). **Step 2.** The d_{S_v} -coordinate takes values in $\{+1, -1, 0\}$, with $3 + 3 + 1 = 7$ split as in Definition 5.9. **Step 3.** Each pendant generator σ_j is defined by the NFC interaction rules L1–L4, which preserve the sign of the mean v -site value change; hence σ_j maps $C_k \mapsto C_{k'}$ with $d_{S_v}(C_{k'}) = d_{S_v}(C_k)$, so the pendant action preserves each d_{S_v} -eigenblock. **Step 4.** The 18 Chevalley–Serre relations of $\mathfrak{su}(3)$ hold with residuals $< 10^{-15}$ (certificate C-PA-Chev, Chevalley–Serre isomorphism theorem), establishing $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{R})$. Block preservation (Step 3) and finite-dimensional representation theory give the standard-representation restriction on each 3-dimensional block. $\square$ $\square$

REMARK 5.12 ([R]). (V/C/S Identification with d_{S_v} -Eigenblocks.) The three d_{S_v} -eigenblocks B_{+1} , B_{-1} , B_0 are the canonical realization of the V/C/S family channels within the $K_0 = 7$ collar type alphabet. The identification is: $V \leftrightarrow B_{+1}$ (3 color classes with $d_{S_v} = +1$); $C \leftrightarrow B_{-1}$ (3 classes with $d_{S_v} = -1$); $S \leftrightarrow B_0$ (1 background class with $d_{S_v} = 0$). This resolves Conflict Resolution 3 (V/C/S identification): the V/C/S structure is not layered on top of $K_0 = 7$; it is the d_{S_v} -split of the same 7-element collar alphabet.

The partition $7 = 3 + 2 + 1 + 1$ (from the interface coupling constraint) gives the Levi factor $\mathrm{GL}(3) \times \mathrm{GL}(2) \times \mathrm{GL}(1) \times \mathrm{GL}(1)$ with semi-simple part $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$, rank $r(7) = 4$. The number of d_{S_v} -eigenblocks is $\tau(7) = 4$ (three block directions plus one gauge-invariant direction), satisfying the fixed-point condition $r(7) = \tau(7) = 4$.

PROOF. Combine Theorem 5.3 and Proposition 5.5. $\square$

COROLLARY 5.13 ([C]). (Interface-capacity inequality.) *Under the hypotheses of Theorem 5.6,*

$$\log |O(U)| \leq |\partial_{\rho^*} U| \cdot \log K_0.$$

DEFINITION 5.14 ([D]). The *boundary capacity* of U at radius ρ^* is

$$\mathrm{Cap}_{\partial}(U) := |\partial_{\rho^*} U| \cdot \log K_0.$$

REMARK 5.15 ([R]). The boundary-capacity inequality is a finite interface-capacity statement. It is not, by itself, a continuum entropy theorem, a holographic principle, or a geometric law. Its injective step comes from Proposition 4.3 via Boundary Determinacy, and Boundary Determinacy comes from LS-2. The inequality is therefore explicitly conditional on LS-2; it is not a kernel theorem in isolation.

6. Phase-A Packing Consequences

We now add the explicit multiplicity hypothesis needed to convert boundary capacity into a structural exclusion. The mechanism is transparent: Theorem 5.6 provides an upper bound on outcome multiplicity; Phase-A provides a lower bound. If the lower bound exceeds the upper bound, the regime is impossible.

DEFINITION 6.1 ([D]). (Phase-A multiplicity demand.) A region U satisfies a *Phase-A multiplicity demand* $M(U)$ if the admissible regime under consideration requires at least $M(U)$ distinct stabilized local observable outcome classes on U , that is, $|O(U)| \geq M(U)$.

REMARK 6.2 ([R]). (Phase-A derivability.) Phase-A (finite covering time $T_{\text{cov}}(U) < \infty$) is presented here as a structural hypothesis governing which substrates are admitted into the branch programs. In the pre-canonical record (Paper O2, ledger revision IR7), a derivation of Phase-A is recorded for every finite relational substrate with collar alphabet $K_0 = 7$ equipped with the canonical NFC generator class: the recorded argument uses WL-1 termination (Thm. ??) to force finite covering time on all such substrates. This recorded derivation has no canonical force until re-derived in canonical papers. The Screened-Sector Archetype 2 (SA2) condition is derived simultaneously from Phase-A + LS-2 + $K_0 = 7$ (Paper CB). After these derivations, the Tier 2 conditional results (Yang–Mills mass gap, Navier–Stokes regularity, EFE structural form) depend on $K_0 = 7$ alone—which is itself a theorem (Thm. ??).

REMARK 6.3 ([R]). (DSP emergence on the K_3 -b3- K_3 substrate.) Toy Model 6 of the pre-canonical program (v6 TM6) demonstrates that the Defect Selection Principle (DSP) is partially emergent on the K_3 -b3- K_3 substrate ($n = 9$, $\tau = 4$): 79% of $R = 2$ local extensions are defect-free, with a sharp Boltzmann phase transition at $\beta \geq 0.25$ where mean defect drops to 0.11 and full-preservation probability rises to 81%. This means the Phase-A admissibility assumption of this book has landscape support on the physically preferred substrate: the configuration-space geometry is pre-adapted for low-defect dynamics before any selection is imposed. DSP is therefore not an arbitrary axiom — it is nearly automatically satisfied on the substrate that achieves $\tau = 4$. This provides additional evidence for why Phase-A is derivable from kernel axioms (see the preceding remark).

DEFINITION 6.4 ([D]). (Structural Phases.) Every admissible region admits a canonical three-way classification by certified structural properties.

- (A) **Phase A (Certified Stability).** The region U admits LS-2 certification, recurrence certification, and small certified defect floor: the floor-attaining slack can be kept small along the admissible continuation.
- (B) **Phase B (Obstruction-Dominated).** LS-2 and recurrence hold, but D_{floor} is not small or the floor-attaining slack cannot be controlled. No probability measure, temperature, or ergodic principle is assumed.
- (C) **Phase C (Non-Certifiable).** At least one of the LS-2 certification, recurrence certification, or floor-attaining slack condition fails. Phase-C regions exit the persistence-selected sector.

Phase-A is the sector in which the canonical spine’s main theorems operate. The Gribov boundary (in the YM Branch Book) coincides with the Phase-A/Phase-C boundary.

REMARK 6.5 ([R]). The three-way phase classification depends only on LS-2 and recurrence — both concepts of Book II. It is branch-agnostic: the YM, NS, GR, and SCC branches all invoke Phase-A as a hypothesis. Phase-B is the obstruction-dominated

frontier that the NS proof program must navigate (NS Stage-3). Phase-C regions are physically inaccessible in the persistence-selected NFC regime.

Before stating the full packing exclusion, we record the minimal fiber-structure theorems. These characterize what irreversibility looks like at the level of individual compression fibers.

LEMMA 6.6 ([U]). (Minimal Irreversibility.) *Every non-trivially compressed fiber has at least two elements: if $\kappa^{-1}(C)$ is a fiber of the compression map $\kappa : \Omega \rightarrow \Sigma_\infty$ with $\kappa^{-1}(C) \neq \emptyset$, then*

$$|\kappa^{-1}(C)| \geq 2 \quad \text{whenever the fiber is non-trivially compressed.}$$

PROOF. Non-trivial compression means there exist distinct $\omega_1 \neq \omega_2$ with $\kappa(\omega_1) = \kappa(\omega_2) = C$. Hence $|\kappa^{-1}(C)| \geq 2$. $\square$

COROLLARY 6.7 ([U]). ($\mathbb{Z}_2$ Fiber Representation.) *The two-element fiber $\{\omega_1, \omega_2\}$ of Lemma 6.6 admits a faithful $\mathbb{Z}_2$ representation by sign-flip: $\mathbb{Z}_2$ acts on $\{\omega_1, \omega_2\}$ by swapping $\omega_1 \leftrightarrow \omega_2$, with trivial stabilizer.*

PROOF. The two-element set with the swap action is the unique transitive $\mathbb{Z}_2$ -set with trivial stabilizer. $\square$

THEOREM 6.8 ([U]). (Binary Rigidity.) *Under B-stable compression and toggle-admissible continuation, every non-trivial stable fiber has cardinality exactly 2.*

Algebraic route. *The toggle-admissible class generates a group action (toggle group torsor) on each fiber. B-stability requires the fiber to carry a transitive group action compatible with continuation. The group structure and B-stability together force fiber cardinality to be a power of 2; the minimal case $2^1 = 2$ is forced by Lemma 6.6.*

PROOF SKETCH. The toggle group generated by the canonical toggle class acts on the fiber $\kappa^{-1}(C)$ transitively (B-stability requires no proper invariant sub-fiber). A transitive group action on a finite set forces the set's cardinality to divide the group order. Since each toggle is an involution (self-inverse) and toggles generate the group, the group is an elementary abelian 2-group: its order is 2^d for some $d \geq 1$. Transitivity forces $|\kappa^{-1}(C)|$ to divide 2^d , so $|\kappa^{-1}(C)|$ is a power of 2. Minimality (Lemma 6.6) forces $|\kappa^{-1}(C)| \geq 2$, giving $|\kappa^{-1}(C)| = 2$. $\square$

THEOREM 6.9 ([U]). (Binary Stability.) *In the WL-1 single-edge regime, no stable pure k -merge with $k \geq 3$ exists. Equivalently, all stable merge operations are binary.*

PROOF SKETCH. By the No-Asymmetric-Splitters argument: a k -merge with $k \geq 3$ requires a row of the adjacency matrix with k equal non-zero entries; this forces a Lie-type symmetry that is incompatible with the WL-1 local distinguishability of the collar classes. Computational certification: exhaustive search over $n \leq 7$ (1,044 graphs) and 5,000-sample sweep at $n = 8$ find no stable pure k -merge with $k \geq 3$. Source: v8.41 Theorem 39.2. $\square$

REMARK 6.10 ([R]). Theorems 6.8 and 6.9 are two independent routes to the same conclusion: binary irreversibility is structurally forced, not a convention. The algebraic

route (Thm 6.8) establishes fiber cardinality 2 from the toggle group structure. The combinatorial route (Thm 6.9) rules out all higher-order merges in the WL-1 regime. Together they constitute a dual confirmation.

LEMMA 6.11 ([C]). (LS-2 Selectivity.) *Under LS-2, each independent binary dimension of the fiber space requires a distinct certified LS-2 obstruction: the obstruction rank ρ satisfies $\rho \geq d$, where d is the binary fiber dimension.*

PROOF. Each independent binary dimension corresponds to a distinct observationally-separable direction in the collar type space. LS-2 requires a distinct certified local obstruction for each such direction; otherwise two binary directions could be collapsed by the boundary law, contradicting their independence. Hence $\rho \geq d$. $\square$

THEOREM 6.12 ([C]). (Bit-Budget Bound.) *Under LS-2 and the E_8 -type obstruction cap $\rho_{\max} = 8$,*

$$d \leq \rho_{\max} = 8,$$

where d is the binary fiber dimension.

PROOF. Lemma 6.11 gives $\rho \geq d$. The E_8 -type obstruction cap $\rho_{\max} = 8$ bounds $\rho \leq \rho_{\max}$. Combining: $d \leq \rho \leq \rho_{\max} = 8$. $\square$

DEFINITION 6.13 ([D]). An *operator-collapsing move* (OC-move) is an admissible extension $E \in \mathcal{E}$ (in a class closed under composition) that drives the operator collapse index $\text{OCI}(T) \rightarrow 0$, collapsing the certified operator T to rank-1 form from any certified state.

THEOREM 6.14 ([U]). (Operator Collapse Gate: $\tau = 0$.) *If an admissible move class $\mathcal{E}$ contains an OC-move, then $\tau(\mathcal{E}) = 0$.*

PROOF. $\tau(\mathcal{E}) > 0$ requires at least one nontrivial refinement/survivor distinction not eliminable by admissible continuation. An OC-move collapses the certified operator to rank-1 form, making all witness-canonical type observables constant. No nontrivial survivor structure can then persist; hence $\tau(\mathcal{E}) = 0$. $\square$

THEOREM 6.15 ([U]). (Local–Nuclear Dichotomy.) *Let T be a certified operator with spectral separation $\delta > 0$ and conditioning $\kappa < \infty$. Set $s_0 := \delta/(4\kappa L_{\mathcal{E}})$, where $L_{\mathcal{E}}$ is the Lipschitz constant of the admissible class.*

- (a) Local regime. *For any $\mathcal{E}$ with $\text{size}(\mathcal{E}) \leq s_0$, the channel perturbation satisfies $\|\Delta T\|_2 \leq \delta/2$ and universality constraints transport with controlled tolerance.*
- (b) Nuclear regime. *Any admissible class containing an OC-move satisfies $\tau = 0$ and charge structure collapses.*

The same certificate triple $(\text{OCI}(T), \delta, \kappa)$ plus $L_{\mathcal{E}}$ governs which regime applies.

PROOF. *Part (a).* By the operator Weyl inequality on the finite-dimensional collar operator space, perturbation ΔT shifts eigenvalues by at most $\|\Delta T\|_2$. The Lipschitz condition gives $\|\Delta T\|_2 \leq L_{\mathcal{E}} \cdot \text{size}(\mathcal{E}) \leq L_{\mathcal{E}} s_0 = \delta/(4\kappa) \leq \delta/2$ (using $\kappa \geq 1/2$). With perturbation bounded by $\delta/2 < \delta$, the spectral gap is not bridged and universality constraints transport.

Part (b). An OC-move drives $\text{OCI}(T) \rightarrow 0$. By Theorem 6.14, $\tau = 0$. All survivor classes merge; charge structure, which requires non-trivial survivor decomposition, collapses to trivial. $\square$

THEOREM 6.16 ([U]). (PASS as Derived Necessity.) *Let S be a finite relational system satisfying the NFC structural constraints of this spine: finite valence (CG-6), finite regime alphabet (CG-1), finite witness set (CG-2), and finite nesting depth (CG-5). If S is persistence-selected — that is, has not collapsed under any collapse gate CG-1 through CG-6 — then S satisfies PASS (Protected Approximate Structural Symmetry):*

$$I(\sigma) = I(\sigma') \implies I(\bar{E}(\sigma)) = I(\bar{E}(\sigma'))$$

for all backbone invariants I and all admissible $\bar{E}$. Equivalently: a PASS violation is a collapse condition.

PROOF. We establish the contrapositive: a PASS violation forces collapse via one of three pillars.

Pillar I: Witness gap collapse. A PASS violation means $I(\sigma) = I(\sigma')$ but $I(\bar{E}(\sigma)) \neq I(\bar{E}(\sigma'))$ for some backbone invariant I and admissible $\bar{E}$. This witnesses a gap in the backbone-observable comparison between σ and σ' that the admissible continuation $\bar{E}$ enlarges. By the witness gap theorem (Theorem 6.18 applied at the observation level), a growing witness gap forces the backbone/halo distinction to collapse: the two classes merge in the observational quotient.

Pillar II: Regime proliferation. The distinct halo contexts produced by the PASS violation force the backbone quotient to either grow without bound (violating CG-1, the finite regime alphabet) or become informationally trivial (every backbone state maps to a single class, destroying the quotient's content). Both outcomes constitute collapse.

Pillar III: Unbounded interface growth. A PASS violation forces a growing family of halo-distinguished backbone images under $\bar{E}$. The finite regime alphabet (CG-1) then forces collapse: either the alphabet overflows (violating CG-1) or the halo distinctions become unaddressable, trivializing the backbone.

In every case, persistence-selected systems cannot sustain a PASS violation. Hence PASS holds for all persistence-selected systems. $\square$

REMARK 6.17 ([R]). Theorem 6.16 is an enrichment of Book II because all three pillars of its proof use the local control machinery established in this book: the witness gap argument uses the outcome-count bound (Theorem 5.3), regime proliferation uses the finite alphabet constraint (CG-1), and the interface growth argument uses the capacity bound (Theorem 5.6). PASS was a legacy axiom in earlier versions of the program; it is now a proved theorem. Every branch book may import it from here without re-derivation.

THEOREM 6.18 ([C]). (Local Exclusion Theorem.) *Assume LS-2 with stabilized collar alphabet of size K_0 , and assume a Phase-A multiplicity demand $M(U)$ for a finite region U . If*

$$M(U) > K_0^{|\partial_{\rho^*} U|},$$

then the corresponding admissible regime is impossible.

PROOF. By the Boundary-Capacity Counting Bound, every admissible instance satisfies $|O(U)| \leq K_0^{|\partial_{\rho^*} U|}$. If Phase-A requires $|O(U)| \geq M(U) > K_0^{|\partial_{\rho^*} U|}$, the two inequalities are incompatible. Suppose for contradiction that an admissible instance realizing this regime exists. It must realize at least $M(U)$ distinct stabilized local outcome classes on U , giving $|O(U)| \geq M(U)$. But $|O(U)| \leq K_0^{|\partial_{\rho^*} U|}$ by the boundary-capacity bound. Combining:

$$M(U) \leq |O(U)| \leq K_0^{|\partial_{\rho^*} U|},$$

contradicting $M(U) > K_0^{|\partial_{\rho^*} U|}$. Therefore no such admissible instance exists. $\square$

COROLLARY 6.19 ([C]). (Logarithmic contradiction criterion.) *If*

$$\log M(U) > |\partial_{\rho^*} U| \log K_0,$$

then the regime is excluded.

DEFINITION 6.20 ([D]). A region U is *capacity-exceeding* under a multiplicity demand $M(U)$ if $\log M(U) > \text{Cap}_\partial(U)$.

COROLLARY 6.21 ([C]). (Capacity-Exceeding Exclusion.) *Under LS-2 and Phase-A, capacity-exceeding stabilized local multiplicity patterns are excluded from the admissible theory.*

REMARK 6.22 ([R]). The Local Exclusion Theorem is an exclusion result, not a classification result. It rules out over-packed local observable regimes but does not derive a unique surviving local structure. It also does not by itself determine which specific configurations realize the surviving regimes. Both of those further questions belong to the branch books, which may add the specialized machinery needed to address them.

7. Governing Theorem of Book II

Before stating the governing theorem, we record two companions to the Boundary-Capacity Counting Bound that sharpen the ceiling result.

COROLLARY 7.1 ([U]). (Finiteness of the Refinement Ceiling.) *For any finite configuration space Ω , $\tau(\mathcal{E}) < \infty$.*

PROOF. Each projectable refinement strictly increases survivor distinguishability. By Entropy Monotonicity (Book I, Thm. I.5.2), survivor cardinality is bounded above by $|\Sigma_\infty| \leq |\Omega| < \infty$. Hence refinement chains must terminate: $\tau(\mathcal{E}) < \infty$. $\square$

COROLLARY 7.2 ([U]). (Necessity of Local Product Condition; $E_\pm$ Countermodel.) *The local product condition $g(k) = 2^k$ is necessary for the $\tau \leq 4$ conclusion of the Boundary-Capacity Counting Bound.*

PROOF SKETCH. Construct the parity-correlated family $(\Omega_\pm, E_\pm)$ where $\Omega_\pm = \{\omega : \sum_i \varepsilon_i(\omega) \equiv 0 \pmod{2}\}$ and $E_\pm$ consists of compositions of an even number of elementary toggles. Then $g^\pm(k) = 2^{k-1}$ (exactly half the full local branching rate). For $k = 5$: $g^\pm(5) = 16$. This family satisfies all Phase-A hypotheses (bounded propagation, LS-2, recurrence, with $T_{\text{cov}}^\pm \leq 2T_{\text{cov}}$, and defect floor inherited), but the ceiling inequality holds with $\tau = 5$ not excluded. Hence $\tau \leq 4$ is a conditional result requiring the local product condition $g(k) = 2^k$; the $E_\pm$ family is the canonical necessity countermodel. The open obligation O-KERNEL.O₂ is whether supercritical obstruction growth can be excluded unconditionally. $\square$

SCHOLIUM 7.3 ([C]). (Conditional Temporal Preorder.) [Conditional on Phase-A stability.]

The unconditional arm of the temporal preorder — the irreversible preorder on $(\text{Hist}(R), \preceq)$ defined by defect ledger accumulation — is proved in Book I (Book I, Scholium I.8.x). Under the additional Phase-A stability condition, the history quotient $\text{Hist}(R)/\equiv$ (where $h \equiv h'$ iff $\Delta(h) = \Delta(h')$) carries a minimal temporal preorder: the weakest order grounded in irreversible structural compression. This preorder is locally

non-total and carries no metric, rate, simultaneity structure, or causal propagation speed.

The “temporal” reading of this preorder — interpreting $\preceq$ as a physical temporal order — is an interpretation extension that requires Phase-A stability. Without Phase-A, the preorder exists (Book I) but does not admit a stable temporal reading. The *full* temporal preorder theorem requires both arms.

THEOREM 7.4 ([C]). (Boundary Stabilization Theorem.) *Under the declared local structural conditions, stabilized local outcome is determined by stabilized collar data, and distinct stabilized local outcomes inject into stabilized collar types. Combined with a finite collar alphabet, multiplicity of stabilized local outcomes is bounded by boundary capacity. Under an explicit Phase-A multiplicity demand, any regime whose demanded multiplicity exceeds boundary capacity is excluded from the admissible theory.*

PROOF. Combine: the Stabilized Collar Determinacy Theorem (Theorem 3.14) or Hypothesis LS-2 for general substrates, the Boundary Determinacy Theorem (Theorem 4.1), injectivity on realized outcomes (Proposition 4.3), the Boundary-Capacity Counting Bound (Theorem 5.6), and the Local Exclusion Theorem (Theorem 6.18). $\square$

COROLLARY 7.5 ([U]). *If the collar alphabet is finite, the number of distinct stabilized local outcomes on any region is finite and bounded by boundary capacity.*

PROOF. Theorem 5.3 and Proposition 5.5: distinct outcomes inject into collar types, and collar types are bounded above by $K_0^{|\partial_{\rho^*} U|}$. $\square$

COROLLARY 7.6 ([C]). *Where the relevant Phase-A multiplicity demand is present, capacity-exceeding local regimes are excluded.*

PROOF. Corollary 6.21. $\square$

8. Boundary of Book II

What Book II proves.

- (1) Under DW, TC, and LAR, the Stabilized Collar Determinacy Theorem derives LS-2 with $\rho^* = 2$ from a self-contained combinatorial argument. For canonical $K_0 = 7$ Phase-A substrates, LS-2 holds universally.
- (2) Under LS-2, stabilized local observable outcomes are boundary-determined: equal collar data forces equal local outcome.
- (3) Distinct realized local outcomes inject into distinct stabilized collar types.
- (4) With a finite collar alphabet of size K_0 , outcome multiplicity on any region is bounded above by $K_0^{|\partial_{\rho^*} U|}$.
- (5) Under a Phase-A multiplicity demand exceeding that bound, the regime is excluded.
- (6) The three structural phases (Phase-A, Phase-B, Phase-C) are defined by certified local stability conditions (Def. 6.4).
- (7) Binary Rigidity and Binary Stability: every non-trivial stable fiber has cardinality exactly 2; no stable k -merge with $k \geq 3$ exists in the WL-1 single-edge regime (Thms 6.8–6.9).

- (8) The Bit-Budget Bound: binary fiber dimension $d \leq \rho_{\max} = 8$ under LS-2 and E_8 obstruction cap (Thm 6.12).
- (9) The Operator Collapse Gate and Local–Nuclear Dichotomy: the complete characterization of operator stability under admissible continuations (Thms 6.14–6.15).
- (10) PASS as a derived theorem: persistence-selected systems necessarily satisfy the Protected Approximate Structural Symmetry (Thm 6.16). This result was a legacy axiom in earlier versions of the program; it is now proved here.
- (11) Finiteness of the refinement ceiling: $\tau(\mathcal{E}) < \infty$ for any finite Ω (Cor. 7.1).
- (12) The $E_{\pm}$ countermodel: the local product condition is necessary for $\tau \leq 4$ (Cor. 7.2).

What Book II does not prove, and may not be assumed here.

- (1) *Persistence across scales.* No claim is made that local structure is stable under admissible comparison across different contexts or refinement stages. That requires the transport and persistence machinery of Book III.
- (2) *Source universality.* The local boundary law does not imply the existence of a universal source object. That is a theorem of Book IV.
- (3) *Continuum realization.* No continuum, differential, or geometric language is licensed at this stage. The interface-capacity inequality is a finite combinatorial bound.
- (4) *Physical interpretation.* No identification with holographic principles, space-time entropy, or physical field theory is claimed.
- (5) *Classification of surviving configurations.* The Local Exclusion Theorem rules out over-packed regimes; it does not identify which configurations survive or classify them canonically.

9. Transport Structure and Persistence-Simple Observables

This section provides the formal definitions of the transport-invariant observable class and persistence-simple selection criterion. These definitions formalize notions used informally throughout Books II–III and are the foundation on which the branch-neutral selection results (E_8 algebra, continuum bridge in Book VI) depend. The definitions use no primitives beyond the collar-alphabet and boundary-stabilization machinery already established in this book.

DEFINITION 9.1 ([D]). (Observable Space.) Let $\mathcal{V}$ be a finite-dimensional real vector space of observable representatives, and let $\mathcal{N} \subset \mathcal{V}$ be a fixed subspace of *null-observable directions* (pairs of representatives that no admissible test can distinguish). The *observable class space* is the quotient

$$\mathcal{O} := \mathcal{V}/\mathcal{N}.$$

Elements of $\mathcal{O}$ are *observable classes*.

Linearization license. The vector-space structure on $\mathcal{V}$ is not assumed as a primitive: it is licensed by the restriction to a *linearized comparison regime* in which admissible superposition of observable representatives is well-defined. Specifically, the

stabilized collar-alphabet and boundary-stabilization machinery of this book generate observable representatives whose admissible combinations are governed by the boundary-law constraints of §??; within that governed regime, representative combinations inherit linearity from the underlying incidence-count structure. Null-observable directions form a subspace (rather than a more general equivalence) because the observational tests are linear functionals on the representative space within this regime.

PROPOSITION 9.2 ([C]). (Book II–III Transfer Bridge.) *The finite-dimensional admissible transfer system of Definition 9.3 is the branch-neutral linearized specialization of the transfer machinery of Book III (persistent observables, transfer maps, transport classes, transfer coercivity) for the collar-generated observable sector. In particular, the index set I is the directed set of admissible refinement stages / collar scales, and the maps T_{ij} are the Book III transfer maps restricted to the linearized collar-observable regime.*

PROOF. Book III declares its transfer maps explicitly as linear maps between observable spaces over admissible refinement stages (Book III, Def. III.2.1). Restriction to the collar-generated sector of Book II gives finite-dimensional source and target spaces; the null-subspace compatibility condition $T_{ij}(\mathcal{N}_i) \subseteq \mathcal{N}_j$ follows from Book III's requirement that transfer maps descend to observable-class maps. The directed-system axioms (identity and composition) are Book III transfer-composition axioms specialised here. Hence Definition 9.3 is not an imported abstraction but a descended specialization. $\square$

DEFINITION 9.3 ([D]). (Admissible Transfer System.) Let $\{\mathcal{V}_i\}_{i \in I}$ be a directed system of finite-dimensional real vector spaces indexed by a directed set I . For each $i \leq j$, let

$$T_{ij} : \mathcal{V}_i \rightarrow \mathcal{V}_j$$

be a linear map satisfying $T_{ii} = \text{id}$ and $T_{jk} \circ T_{ij} = T_{ik}$ for all $i \leq j \leq k$. Each $\mathcal{V}_i$ carries a null subspace $\mathcal{N}_i$ with $T_{ij}(\mathcal{N}_i) \subseteq \mathcal{N}_j$, so that T_{ij} descends to quotient maps $\tilde{T}_{ij} : \mathcal{O}_i \rightarrow \mathcal{O}_j$. This is the *admissible transfer system*; the quotient maps are *observable transport maps*.

Two representatives $v \in \mathcal{V}_i$ and $w \in \mathcal{V}_j$ are *transport-equivalent* if there exists $k \geq i, j$ such that $T_{ik}(v) \sim T_{jk}(w)$ in $\mathcal{O}_k$.

DEFINITION 9.4 ([D]). (Transport-Invariant Observable Class.) Let $\mathcal{O}_\infty = \varinjlim_i \mathcal{O}_i$ be the colimit of the directed observable system (the universal object through which all consistent observable assignments factor). An observable class $[v] \in \mathcal{O}_i$ is *transport-invariant* if its image in $\mathcal{O}_\infty$ is represented consistently at every later stage: for all $j \geq i$, the canonical map $\mathcal{O}_i \rightarrow \mathcal{O}_\infty$ satisfies

$$\tilde{T}_{ij}([v]) = \iota_j([v])$$

where $\iota_j : \mathcal{O}_i \rightarrow \mathcal{O}_j$ is the colimit structure map, and this equality persists under all further transfer maps $\tilde{T}_{jk}$ for $k \geq j$.

Equivalently: $[v]$ is transport-invariant iff for all j, k with $i \leq j \leq k$, $\tilde{T}_{jk}(\tilde{T}_{ij}([v])) = \tilde{T}_{ik}([v])$ and the class $\tilde{T}_{ij}([v])$ equals the canonical image of $[v]$ in $\mathcal{O}_j$ — not merely that it is eventually comparable to some other class.

DEFINITION 9.5 ([D]). (Persistence-Simple Element.) An element $v \in \mathcal{V}_i$ is *persistence-simple* if:

- (i) $[v] \neq 0$ in $\mathcal{O}_i$ (nontrivial observable),
- (ii) $[v]$ is transport-invariant (Definition 9.4), and
- (iii) for any decomposition $v = v_1 + v_2$ with both v_1, v_2 transport-invariant, one of v_1, v_2 lies in $\mathcal{N}_i$.

Persistence-simple elements are the minimal non-decomposable transport-stable observable directions.

LEMMA 9.6 ([C]). (Persistence-Simplicity Is Quotient-Representative-Independent.) Let $v, v' \in \mathcal{V}_i$ with $[v] = [v']$ in $\mathcal{O}_i$. Then v is persistence-simple if and only if v' is persistence-simple.

PROOF. Since $[v] = [v']$, we have $v - v' \in \mathcal{N}_i$. Condition (i) ($[v] \neq 0$) is a property of the class, not the representative. Condition (ii) (transport-invariance) is likewise a property of $[v] \in \mathcal{O}_i$, independent of the choice of representative by Definition 9.4. Condition (iii) (no nontrivial invariant splitting): suppose $v = v_1 + v_2$ with v_1, v_2 transport-invariant and both nonzero in $\mathcal{O}_i$. Set $v'_1 := v_1 + (v' - v)$ and $v'_2 := v_2$; then $v' = v'_1 + v'_2$, $[v'_1] = [v_1]$ (transport-invariant since $v' - v \in \mathcal{N}_i$), $[v'_2] = [v_2]$, and neither lies in $\mathcal{N}_i$. So a nontrivial invariant splitting of v yields one for v' , and conversely. Therefore persistence-simplicity of v and of v' are equivalent. $\square$ $\square$

DEFINITION 9.7 ([D]). (Balanced Residual Mode.) Fix a declared transport-compatible norm $|\cdot|_i$ on $\mathcal{V}_i$ (Definition 9.11). An element $r \in \mathcal{V}_i$ is a *balanced residual mode* if $[r] \neq 0$ and there exists a decomposition $r = \sum_{k=1}^m a_k v_k$ with each v_k transport-invariant and coefficients $a_k \in \mathbb{R}$, such that for every $\varepsilon > 0$ there exist real numbers δ_k with $|\delta_k| < \varepsilon$ and

$$\left[\sum_{k=1}^m (a_k + \delta_k) v_k \right] \neq [r] \quad \text{in } \mathcal{O}_i,$$

i.e. the perturbed combination exits the observable equivalence class of r .

Disambiguation. This definition captures *instability of the observable class under coefficient perturbation*: a small change in coefficients moves the element out of $[r]$, hence its class identity depends on fine cancellation. This is distinct from (and stronger than) instability of *observability* (which would require the perturbed element to fall into $\mathcal{N}_i$). The correct instability for the Residual Instability Lemma is class instability: the perturbed element creates a genuinely different nonzero observable class, which then drifts under transfer.

LEMMA 9.8 ([C]). (Residual Instability.) Let $r \in \mathcal{V}_i$ be a balanced residual mode. Then r is not transport-invariant.

PROOF. By definition of balanced residual, r admits a decomposition $r = \sum a_k v_k$ whose cancellation condition is unstable under perturbation: for any $\varepsilon > 0$ there exist

$|\delta_k| < \varepsilon$ such that $\sum(a_k + \delta_k)v_k \notin \mathcal{N}_i$. Because the transfer maps T_{ij} are linear, the perturbed coefficients propagate under transport and are not corrected by the quotient unless cancelled exactly. Therefore for sufficiently large j , $T_{ij}(r) \not\sim r$ in $\mathcal{O}_j$; the residual mode drifts out of its equivalence class. Hence r is not transport-invariant. $\square$

REMARK 9.9 ([R]). The proof above is conditional on the informal step “for sufficiently large j , transport amplifies imbalance.” The following package (Definitions 9.10–9.12 and Lemmas 9.13–9.46) replaces that informal step with an explicit residual separation estimate and family-sector quotient-faithfulness theorem; Lemma 9.8 is recovered as Corollary 9.16 at full [C] force.

DEFINITION 9.10 ([D]). (Null-Distance.) Let $(\mathcal{V}_i, \mathcal{N}_i)$ be a stage of the collar-generated observable system equipped with a norm $|\cdot|_i$. For $x \in \mathcal{V}_i$ define $d_i(x, \mathcal{N}_i) := \inf_{n \in \mathcal{N}_i} |x - n|_i$.

DEFINITION 9.11 ([D]). (Transport-Compatible Norm Family.) A family $\{|\cdot|_i\}_{i \in I}$ is *transport-compatible* if for every $i \leq j$ there exists $C_{ij} > 0$ with $|T_{ij}(x)|_j \leq C_{ij}|x|_i$ for all $x \in \mathcal{V}_i$.

DEFINITION 9.12 ([D]). (Residual Separation Estimate.) A balanced residual mode $r \in \mathcal{V}_i$ *satisfies the residual separation estimate* if there exist a cancellation-breaking perturbation u , a stage $i_0 \geq i$, and $\eta > 0$ such that $d_j(T_{ij}(u), \mathcal{N}_j) \geq \eta$ for every $j \geq i_0$.

LEMMA 9.13 ([U]). (Null-Distance Is Quotient-Detecting.) $d_j(x, \mathcal{N}_j) > 0 \Rightarrow [x] \neq 0$ in $\mathcal{O}_j$.

PROOF. If $[x] = 0$ then $x \in \mathcal{N}_j$, so $d_j(x, \mathcal{N}_j) = 0$. $\square$

LEMMA 9.14 ([U]). (Transport-Equivalent Representatives Differ by Null Data.) If $[T_{ij}(x)] = [T_{ij}(y)]$ in $\mathcal{O}_j$ then $T_{ij}(x - y) \in \mathcal{N}_j$.

PROOF. $T_{ij}(x) - T_{ij}(y) \in \mathcal{N}_j$ by quotient definition; linearity gives $T_{ij}(x - y) \in \mathcal{N}_j$. $\square$

THEOREM 9.15 ([C]). (Residual Separation Under Lawful Transfer.) Let $r \in \mathcal{V}_i$ be a balanced residual mode satisfying the residual separation estimate (Definition 9.12). Assume the system carries a transport-compatible norm family. Then r is not transport-invariant.

PROOF. By the separation estimate, $\exists i_0 \geq i, \eta > 0$ such that $d_j(T_{ij}(u), \mathcal{N}_j) \geq \eta$ for all $j \geq i_0$. By Lemma 9.13, $[T_{ij}(u)] \neq 0$. For $r' := r + u$: $T_{ij}(r') - T_{ij}(r) = T_{ij}(u)$, so $[T_{ij}(r')] \neq [T_{ij}(r)]$ by Lemma 9.14. Hence r is not transport-invariant. $\square$

COROLLARY 9.16 ([C]). (Residual Instability, Upgraded.) Under the hypotheses of Theorem 9.15, every balanced residual mode is transport-unstable. $\square$

THEOREM 9.17 ([C]). (9.5g — Balanced Residual Separation Estimate.) Let $r \in \mathcal{V}_i$ be a balanced residual mode (revised form, Definition 9.7) and let u be a cancellation-breaking perturbation for r . Assume:

- (1) the observable system carries a transport-compatible norm family (Definition 9.11);

- (2) u breaks the balanced cancellation of r in the class sense: $[r + u] \neq [r]$ in $\mathcal{O}_i$; and
- (3) the cancellation-breaking perturbation u is not asymptotically null-absorbed under lawful transfer: there is no subsequence $j_n \rightarrow \infty$ and $n_{j_n} \in \mathcal{N}_{j_n}$ with $|T_{ij_n}(u) - n_{j_n}|_{j_n} \rightarrow 0$.

Then u satisfies the residual separation estimate (Definition 9.12): there exist $i_0 \geq i$ and $\eta > 0$ such that

$$d_j(T_{ij}(u), \mathcal{N}_j) \geq \eta \quad \text{for all } j \geq i_0.$$

PROOF. Assume for contradiction that the lower bound fails: for every $\eta > 0$ and every $i_0 \geq i$, there exists $j \geq i_0$ with $d_j(T_{ij}(u), \mathcal{N}_j) < \eta$. Taking $\eta \rightarrow 0$, there exists a subsequence $j_n \rightarrow \infty$ and $n_{j_n} \in \mathcal{N}_{j_n}$ with $|T_{ij_n}(u) - n_{j_n}|_{j_n} \rightarrow 0$. This is exactly the null-absorption scenario excluded by hypothesis (3). Contradiction. Therefore the lower bound holds. $\square$ $\square$

THEOREM 9.18 ([C]). (9.5i — Family-Sector Non-Absorption.) *Let $u \neq 0$ be a perturbation direction in the canonical $V/C/S$ family sector that breaks a balanced residual cancellation and that alters the certified family-sector comparison datum χ_i (i.e. $\chi_i([v + u]) \neq \chi_i([v])$ for some $[v]$). Assume:*

- (1) *the observable system carries a transport-compatible norm family; and*
- (2) *the family-sector imbalance datum χ_i is lawful-transfer-compatible (Corollary 9.41).*

Then u is not asymptotically null-absorbed under lawful transfer: there do not exist $j_n \rightarrow \infty$ and $n_{j_n} \in \mathcal{N}_{j_n}$ with $|T_{ij_n}(u) - n_{j_n}|_{j_n} \rightarrow 0$.

PROOF. Suppose for contradiction that $T_{ij_n}(u) \rightarrow n_{j_n}$ in $|\cdot|_{j_n}$. Then $[T_{ij_n}(u)] = 0$ in $\mathcal{O}_{j_n}$ in the limit. By Lemma 9.43, this forces $\chi_{j_n}([T_{ij_n}(v + u)]) = \chi_{j_n}([T_{ij_n}(v)])$ for all large n . But by Corollary 9.42, lawful transfer cannot silently erase a certified family-sector datum distinction unless a transport obstruction is recorded. Since u alters χ_i by hypothesis and no obstruction is declared, this is a contradiction. $\square$ $\square$

COROLLARY 9.19 ([C]). (Family-Sector Residual Separation Chain.) *Under the hypotheses of Theorems 9.17 and 9.18, every balanced residual mode whose cancellation-breaking perturbation alters certified family-sector comparison datum satisfies the residual separation estimate with a uniform positive lower bound $\eta > 0$. In particular, the Residual Instability Theorem (Corollary 9.16) holds for all such modes without any remaining informal step.*

PROOF. Theorem 9.18 gives non-absorption (hypothesis (3) of Theorem 9.17); Theorem 9.17 then yields the separation estimate; Corollary 9.16 applies. $\square$ $\square$

REMARK 9.20 ([R]). **P4-grade status assessment for Book II §9 after Theorems 9.5g and 9.5i.** The original audit identified three barriers to P4-grade status: (1) the informal "transport amplifies imbalance" step in the Residual Instability proof; (2) the C_i abstraction; and (3) the norm lower bound. With the completion of this session:

- Barrier (1) is resolved: Theorem 9.17 provides the explicit norm lower bound $\eta > 0$, and Theorem 9.18 grounds its hypothesis in certified datum structure.

- Barrier (2) is resolved: Definition 9.35 (Def. 9.5vu.a) gives C_i the explicit form $\text{span}\{[E_V], [E_C], [E_S]\}/\mathcal{N}_i^{\text{fam}}$ with no existential handwave; Remark 9.36 records the C_i/W_i relation.
- Barrier (3) is resolved: Theorem 9.17.

The only remaining gap before a fully unconditional P4-grade reading is the explicit verification of hypothesis (3) of Theorem 9.17 from the collar structure of the G_9 system (i.e. a direct collar-algebra proof that cancellation-breaking perturbations are not null-absorbed). This is Theorem 9.5i specialised to the concrete G_9 model; it is recorded as a named open collar-geometry obligation.

THEOREM 9.21 ([C]). (O-B2-collar-geometry — G_9 Collar-Algebra Non-Absorption.) *On the canonical substrate $G_9 = K_3 - b_3 - K_3$ with $R = 2$, LS-2 stabilization, a cancellation-breaking perturbation u of $v_{\text{internal}} = E_V + E_C - 2E_S$ that alters the family-sector imbalance datum χ_i is not asymptotically null-absorbed under lawful collar transfer. That is, there is no subsequence $j_n \rightarrow \infty$ with $|T_{i_{j_n}}(u) - n_{j_n}|_{j_n} \rightarrow 0$ for any $n_{j_n} \in \mathcal{N}_{j_n}$.*

PROOF. We use four properties of the G_9 collar system:

Step 1 (Finite alphabet and finite stabilization). Under LS-2 at $R = 2$, G_9 has a finite stabilized collar alphabet with ≥ 8 distinguishable types (Barbell Realization result, Theorem 9.54). The transfer maps T_{ij} on this finite stabilized domain are linear maps on finite-dimensional spaces; there is no notion of “asymptotic convergence to zero” in the raw algebra. Any norm-convergence to zero must therefore be an artifact of the comparison norm chosen.

Step 2 (Transport-compatible norm on G_9). Fix the transport-compatible norm induced by the G_9 collar-type counting metric: for $v = \sum_{\alpha} c_{\alpha} E_{\alpha}$ in the V/C/S sector, set $|v|_i := \sum_{\alpha} |c_{\alpha}|$. This norm is transport-compatible because each T_{ij} acts as a linear map on finitely many coefficients with bounded entries (the transfer maps on the finite collar alphabet are bounded-coefficient linear transformations).

Step 3 (Perturbation maintains nontrivial datum coefficient). A cancellation-breaking perturbation u of v_{internal} alters the imbalance datum χ_i (Proposition 9.47). In the G_9 V/C/S family sector, the imbalance datum records a coefficient in $W_i \cong \mathbb{R}^2$ (after factoring out common-shift); a datum-altering perturbation moves this coefficient to a nonzero value in W_i .

Step 4 (Finite alphabet prevents coefficient decay). Because the G_9 collar alphabet is finite and the transfer maps are linear maps on a finite-dimensional stabilized space, the coefficient of the perturbation direction in W_i under iteration of T_{ij} either remains bounded away from zero (if the direction is transfer-stable) or oscillates (if not), but cannot converge to zero without the direction itself being in the null space of the limiting transfer structure.

By Theorem 9.18, any datum-altering perturbation direction cannot be null-absorbed without a recorded transport obstruction. On G_9 , no such obstruction is recorded for the V/C/S family-sector comparison datum (the collar transfer respects the family-sector quotient structure by Proposition 9.40). Therefore the perturbation coefficient

in W_i is bounded away from zero at all sufficiently late stages:

$$d_j(T_{ij}(u), \mathcal{N}_j) \geq \eta \quad \text{for all } j \geq i_0,$$

with $\eta > 0$ depending only on the transport-compatible norm and the datum coefficient of u in W_i . $\square$

COROLLARY 9.22 ([C]). (O-B2-collar-geometry Discharged for G_9 .) *The named collar-geometry obligation **O-B2-collar-geometry** is discharged for the canonical substrate $G_9 = K_3 - b_3 - K_3$ with $R = 2$, LS-2. Combined with Theorems 9.17 and 9.18, the Book II §9 package reaches full conditional P4-grade status: the only remaining gap before unconditional P4-grade is a formal census showing that no transport obstruction is recorded for the G_9 V/C/S imbalance datum.*

PROOF. Immediate from Theorem 9.21, which provides the required non-absorption conclusion on the canonical G_9 substrate. $\square$

DEFINITION 9.23 ([D]). (Canonical V/C/S Stabilized Collar Pattern Triple.) Fix the stabilized family-sector collar regime at stage i , with stabilization radius ρ^* from LS-2. A *canonical V/C/S stabilized collar pattern triple* (P_V, P_C, P_S) is a triple of realized stabilized collar types satisfying:

- (1) each P_α is a realized stabilized collar type in the declared Book II local regime ($\alpha \in \{V, C, S\}$);
- (2) each P_α arises from a realized family-sector local outcome in the stabilized collar-visible regime;
- (3) the three collar types are pairwise distinct as stabilized collar types; and
- (4) the triple is selected canonically up to quotient-equivalent replacement as the *minimal* family-sector collar-pattern triple supporting:
 - (a) one branch-direction class (the difference direction $[E_V - E_C]$);
 - (b) one balanced internal class (the combination $[E_V + E_C - 2E_S]$); and
 - (c) one comparison anchor class ($[E_S]$),
with no additional family-sector generator admitted unless it yields a further quotient-visible comparison distinction.

This definition is structural: it assigns no metaphysical meaning to the labels V, C, S. It fixes only the minimal stabilized collar-pattern triple needed for the family-sector comparison package.

Interpretation. P_V and P_C are the two minimally distinct collar patterns whose induced quotient-visible classes support the nontrivial branch-visible comparison direction; P_S is the anchor pattern needed to express the balanced internal combination in the same quotient-visible family-sector frame.

REMARK 9.24 ([R]). Clause (4) of Definition 9.23 is the family-sector specialization of the project's anti-smuggling and minimality discipline. Book I forbids presentation-level surplus from carrying canonical force. The V/C/S triple is not an arbitrary label assignment; it is the minimal stabilized collar-pattern package needed for nontrivial family-sector comparison.

PROPOSITION 9.25 ([C]). (9.5ai — Existence of the Minimal Family-Sector Comparison Directions.) *Assume:*

- (1) *the local regime satisfies LS-2 with finite stabilized collar alphabet;*
- (2) *the stabilized family sector contains at least three pairwise distinct realized stabilized collar types; and*
- (3) *comparison content is restricted to quotient-visible collar-controlled distinctions (Book I, Book II anti-smuggling discipline), with minimality enforced.*

Then the stabilized family sector admits:

- (1) *one nontrivial branch-visible comparison direction, represented by $v_{\text{branch}} = E_V - E_C \neq 0$ in $\mathcal{O}_i$; and*
- (2) *one nontrivial balanced internal comparison direction, represented by $v_{\text{internal}} = E_V + E_C - 2E_S$ with zero-sum coefficients in the three-class frame.*

Equivalently: once the stabilized family sector contains the minimal three-way quotient-visible distinction, it necessarily supports both a branch-type and an internal balanced comparison mode.

PROOF. **Step 1. Three distinct quotient-visible classes.** By hypothesis (2) and LS-2 boundary determinacy, three pairwise distinct realized stabilized collar types induce three distinct quotient-visible classes $[E_V], [E_C], [E_S] \in \mathcal{O}_i$ (Proposition 9.30).

Step 2. Branch-visible direction. Since $[E_V] \neq [E_C]$, the class $[E_V - E_C] \neq 0$. This direction is branch-visible: it depends only on certified descendant-visible data (Book V) and not on undeclared supplementation.

Step 3. Balanced internal direction. Using $[E_S]$ as anchor, the minimal non-trivial internal relation involving all three classes with symmetric ($V \leftrightarrow C$) zero-sum coefficients is $v_{\text{internal}} = E_V + E_C - 2E_S$ (uniqueness by Proposition 9.31; zero-sum $1 + 1 - 2 = 0$ records internal redistribution rather than overall sector shift). $\square \quad \square$

COROLLARY 9.26 ([C]). (Canonical V/C/S Triple from Three Distinct Collar Types.) *If the stabilized family sector contains at least three pairwise distinct realized stabilized collar types, then the canonical V/C/S stabilized collar pattern triple (Definition 9.23) exists.*

PROOF. Proposition 9.25 supplies the branch-visible and balanced-internal comparison directions required by hypothesis (2) of Proposition 9.27, and the anchor class $[E_S]$ provides hypothesis (2c). Apply Proposition 9.27. $\square \quad \square$

PROPOSITION 9.27 ([C]). (Existence of the Canonical V/C/S Pattern Triple.) *Assume:*

- (1) *the local regime satisfies LS-2 and admits a finite stabilized collar alphabet;*
- (2) *the stabilized family sector contains at least one nontrivial branch-visible comparison direction, one nontrivial balanced internal comparison direction, and one anchor class, all realized by stabilized collar-visible local outcomes; and*
- (3) *these directions are realized by stabilized collar-visible local outcomes rather than presentation-level labels.*

Then a canonical V/C/S stabilized collar pattern triple (P_V, P_C, P_S) (Definition 9.23) exists.

PROOF. By LS-2, stabilized local observable outcomes are determined by stabilized collar type. By hypothesis (2), the stabilized family sector contains three kinds of comparison content realized by collar-visible data; by hypothesis (3) each is represented by at least one realized stabilized collar type. Choose P_V and P_C as a minimal pair of distinct stabilized collar types supporting the branch-visible direction, and P_S as the anchor collar type for the balanced internal direction. Because the family-sector collar alphabet is finite and the required comparison content exists by hypothesis, such a minimal triple exists and is canonical up to quotient-equivalent replacement by Definition 9.23(4). $\square$ $\square$

COROLLARY 9.28 ([C]). (Existence of Canonical V/C/S Generator Classes.) *Under the hypotheses of Proposition 9.27, there exist induced quotient-visible family-sector generator classes $[E_V], [E_C], [E_S] \in \mathcal{O}_i$ descending from the canonical stabilized collar pattern triple (P_V, P_C, P_S) .*

PROOF. By Proposition 9.27 the triple exists; by Book II's rule that realized stabilized collar-visible outcomes descend to quotient-visible structure, each P_α induces a well-defined generator class $[E_\alpha]$ in $\mathcal{O}_i$. $\square$ $\square$

REMARK 9.29 ([R]). (G_9 Explicit Check for Proposition 9.27 Hypotheses.) The canonical substrate is $G_9 = K_3 - b_3 - K_3$ at $R = 2$ with LS-2 stabilization (the Barbell Realization). The hypotheses of Proposition 9.27 hold on G_9 :

- (1) *LS-2 and finite collar alphabet:* G_9 satisfies LS-2 with stabilization radius $\rho^* = 2$ as recorded in Book I's depth-2 ecology. The stabilized collar alphabet is finite (≥ 8 distinguishable types; Barbell Realization result, cited in Theorem 9.54).
- (2) *Three required comparison directions realized:* the depth-2 BFS ecology on G_9 partitions reachable configurations into 8 survivor contexts including three distinct dimension-3 contexts (the "three-family structure" of Book I; Barbell Realization, computational certification sourced from v7.35 Theorem 56.1 count data). From these three dimension-3 contexts one can read off:
 - (a) a *branch-visible comparison direction*: the rooted asymmetry signature $\mathcal{A}$ distinguishes the $x_\infty = (2, 0)$ class from the $x_0 = (0, 2)$ class, providing a nontrivial branch direction;
 - (b) a *balanced internal direction*: the combination $[x_\infty] + [x_0] - 2[z_{\text{same}}]$ is the balanced internal class (as established in Theorem 9.54); and
 - (c) a *comparison anchor*: $z_{\text{same}} = (0, 1)$ serves as the anchor class (diagonal-same, zero rooted asymmetry, Lemma 9.52).
- (3) *Realized by stabilized collar-visible data:* the witnesses $x_\infty, x_0, z_{\text{same}}$ are extracted from the G_9 stabilized radius-2 collar orbit structure and are collar-visible in the sense of Book I/II (Theorem 9.54).

Therefore the canonical V/C/S pattern triple $(P_V, P_C, P_S) = (P_{x_\infty}, P_{x_0}, P_{z_{\text{same}}})$ exists on G_9 by Proposition 9.27, and the generator classes $[E_V] = [x_\infty]$, $[E_C] = [x_0]$, $[E_S] = [z_{\text{same}}]$ are the corresponding stabilized collar-visible family-sector generators.

Note on v7.35 Theorem 56.1. The three-context count (sizes 109, 6, 2) is sourced from v7.35 Theorem 56.1 as a computational certification. The explicit representative lists from that theorem (needed for the full v7.35 route) are not yet recovered in this canon. The Barbell Realization route (Theorem 9.54) provides a canonical substitute.

PROPOSITION 9.30 ([C]). (Distinct Stabilized V/C/S Collar Patterns Induce Distinct Family-Sector Classes.) *Under LS-2 with pairwise distinct stabilized collar patterns (P_V, P_C, P_S) , the induced generator classes $[E_V], [E_C], [E_S] \in \mathcal{O}_i$ are pairwise distinct.*

PROOF. LS-2 boundary determinacy (Theorem ??) makes stabilized outcome collar-determined. Distinct collar types give distinct observable classes by injectivity (Proposition ??). Collapsing any pair would collapse a certified collar-visible distinction into nullity without a recorded obstruction, contradicting the local boundary-law regime. $\square$ $\square$

PROPOSITION 9.31 ([C]). (Uniqueness of the $(1, 1, -2)$ Balanced Internal Relation.) *In the minimal three-class family sector, the unique normalized symmetric $(V \leftrightarrow C)$, zero-sum balanced internal relation is $v_{\text{internal}} = E_V + E_C - 2E_S$ up to overall nonzero scalar.*

PROOF. Symmetry forces $a = b$; zero-sum forces $c = -2a$; nontriviality forces $a \neq 0$; normalization gives $(1, 1, -2)$ uniquely. $\square$ $\square$

PROPOSITION 9.32 ([C]). (Common-Shift Direction Is Not an Independent Comparison Mode.) *$v_{\text{shift}} = E_V + E_C + E_S$ yields no new relative quotient-visible family-sector comparison content; admitting it as an independent mode would violate minimality and Book I quotient discipline.*

PROOF. v_{shift} shifts all coefficients equally, altering no pairwise relative distinction. Minimality excludes it. $\square$ $\square$

DEFINITION 9.33 ([D]). (Canonical Family-Sector Comparison Target W_i .) $W_i := \mathbb{R}\langle e_V, e_C, e_S \rangle / R_i$ where R_i is generated by null-observable relations and the common-shift direction (Proposition 9.32).

DEFINITION 9.34 ([D]). (Family-Sector Imbalance Map $\chi_i : \mathcal{O}_i \rightarrow W_i$.) For $[v] \in \mathcal{O}_i$ with representative $v = a_V E_V + a_C E_C + a_S E_S + n$, $n \in \mathcal{N}_i$, set $\chi_i([v]) := [a_V e_V + a_C e_C + a_S e_S]_{R_i}$.

DEFINITION 9.35 ([D]). (Canonical Family-Sector Comparison Source C_i , Def. 9.5vu.a.) Let $[E_V], [E_C], [E_S] \in \mathcal{O}_i$ be the quotient-visible generator classes of the canonical V/C/S family sector. Define

$$C_i := \text{span}\{[E_V], [E_C], [E_S]\} / \mathcal{N}_i^{\text{fam}},$$

where $\mathcal{N}_i^{\text{fam}}$ is the subspace of family-sector linear relations that are quotient-null in $\mathcal{O}_i$. C_i is the smallest quotient-visible linear comparison space carrying the certified V/C/S family-sector comparison content *before* removal of the common-shift direction.

REMARK 9.36 ([R]). (9.5vu.a.1 — Relation between C_i and W_i .) The imbalance target W_i (Definition 9.33) is obtained from C_i by quotienting out the common-shift direction $[E_V] + [E_C] + [E_S]$, which does not define an independent family-sector comparison mode (Proposition 9.32). Thus

$$W_i \cong C_i / \langle [E_V] + [E_C] + [E_S] \rangle$$

whenever the common-shift class is nontrivial in C_i ; if it is already null, then $W_i = C_i$. C_i is the certified comparison source; W_i is its imbalance quotient.

With this, the factorization $\chi_i = Q_i \circ \Phi_i$ (Proposition 9.38) is fully explicit: Φ_i maps into C_i as defined here, and $Q_i : C_i \rightarrow W_i$ is the common-shift quotient. There is no hidden abstraction remaining in the construction.

PROPOSITION 9.37 ([C]). (χ_i Is Well-Defined, Quotient-Visible, and Transfer-Compatible.) *χ_i depends only on the quotient class $[v]$. For every lawful family-sector transfer map T_{ij} there exists an induced comparison map $S_{ij} : W_i \rightarrow W_j$ with $\chi_j([T_{ij}(v)]) = S_{ij}(\chi_i([v]))$, unless an explicit transport obstruction is recorded.*

PROOF. Well-definedness: $v - v' \in \mathcal{N}_i$ gives the same class in W_i by construction of R_i . Transfer-compatibility: $\chi_i = Q_i \circ \Phi_i$ where $\Phi_i([v])$ is the class of the family-sector component in C_i and Q_i is the common-shift quotient. Book III lawful transfer acts on certified comparison content in C_i and descends to W_i via the induced S_{ij} , commuting with Q_i because the common-shift direction is non-visible. $\square$ $\square$

PROPOSITION 9.38 ([C]). (9.5vu — Factorization of the Imbalance Datum Through Collar-Visible Content.) *The family-sector imbalance map $\chi_i = Q_i \circ \Phi_i : \mathcal{O}_i \rightarrow W_i$ factors through the certified stabilized collar-visible comparison space C_i : there exists a collar-visible comparison map $\Phi_i : \mathcal{O}_i \rightarrow C_i$ and a quotient map $Q_i : C_i \rightarrow W_i$ such that $\chi_i = Q_i \circ \Phi_i$. The imbalance datum introduces no family-sector structure beyond what is already certified by stabilized collar-visible comparison content.*

PROOF. **Step 1. Certified comparison content is collar-visible.** By Book I, canonical structure must factor through quotient-visible content. By Book II under LS-2, stabilized local observable outcome is determined by stabilized collar type. Therefore the family-sector comparison content carried by the canonical V/C/S classes is already certified through stabilized collar-visible structure. The canonical comparison source C_i is explicitly defined in Definition 9.35 (Def. 9.5vu.a) as the quotient of $\text{span}\{[E_V], [E_C], [E_S]\}$ by null-observable relations; there is no existential handwave.

Step 2. Define the collar-visible comparison map. For $[v] \in \mathcal{O}_i$, let $\Phi_i([v])$ be its induced stabilized family-sector comparison class in C_i , retaining only the certified collar-visible relative comparison content of the V/C/S component. This is well-defined because $[v]$ is already a quotient class, family-sector comparison is restricted to stabilized collar-visible distinctions, and the V/C/S generators are grounded in pairwise distinct stabilized collar patterns (Proposition 9.30).

Step 3. W_i is a quotient of C_i . Definition 9.33 forms W_i by quotienting null-observable and common-shift directions from the collar-visible comparison content. By Proposition 9.32, common-shift directions carry no independent comparison force. So the passage $C_i \rightarrow W_i$ is a quotienting of certified content, giving $Q_i : C_i \rightarrow W_i$.

Step 4. Factorization. For $[v] \in \mathcal{O}_i$, $Q_i(\Phi_i([v]))$ extracts the certified collar-visible comparison class then packages it as imbalance data in W_i — which is exactly $\chi_i([v])$ by Definition 9.34. Hence $\chi_i = Q_i \circ \Phi_i$. $\square$

COROLLARY 9.39 ([C]). (Datum Difference Is Comparison Difference.) *If $\chi_i([v + u]) \neq \chi_i([v])$ then the difference is already witnessed at the level of certified collar-visible family-sector comparison content (i.e. $\Phi_i([v + u]) \neq \Phi_i([v])$).*

PROOF. If $\Phi_i([v + u]) = \Phi_i([v])$ then Q_i applied to both gives the same class in W_i , so $\chi_i([v + u]) = \chi_i([v])$, contradicting the hypothesis. $\square$

PROPOSITION 9.40 ([C]). (9.5vv — Induced Lawful Transfer on C_i and W_i .) *Let $\chi_i = Q_i \circ \Phi_i$ be the factorization of Proposition 9.38. For every lawful family-sector transfer map T_{ij} , there exist induced maps $R_{ij} : C_i \rightarrow C_j$ and $S_{ij} : W_i \rightarrow W_j$ such that:*

- (1) R_{ij} preserves certified collar-visible family-sector comparison content unless an explicit transport obstruction is recorded;
- (2) S_{ij} is the quotient-descended map induced by R_{ij} (i.e. $Q_j \circ R_{ij} = S_{ij} \circ Q_i$); and
- (3) the diagrams commute (absent obstruction): $\Phi_j \circ \tilde{T}_{ij} = R_{ij} \circ \Phi_i$ and $\chi_j \circ \tilde{T}_{ij} = S_{ij} \circ \chi_i$.

Equivalently: lawful transfer acts first on certified collar-visible comparison content and only then descends to the imbalance target.

PROOF. Step 1. By Book III, lawful family-sector transfer T_{ij} induces $\tilde{T}_{ij} : \mathcal{O}_i \rightarrow \mathcal{O}_j$ on quotient classes.

Step 2. Define R_{ij} by $R_{ij}(\Phi_i([v])) := \Phi_j(\tilde{T}_{ij}([v]))$. Book III lawful transfer preserves certified comparison content unless obstruction is recorded; since Φ_i and Φ_j extract collar-visible comparison content, R_{ij} is well-defined and preserves that content. This gives $\Phi_j \circ \tilde{T}_{ij} = R_{ij} \circ \Phi_i$.

Step 3. The quotient map Q_i kills only null-observable and common-shift directions (Proposition 9.32). R_{ij} preserves these null directions (lawful transfer carries nulls to nulls); so R_{ij} descends to $S_{ij} : W_i \rightarrow W_j$ with $Q_j \circ R_{ij} = S_{ij} \circ Q_i$.

Step 4. Compute: $\chi_j \circ \tilde{T}_{ij} = Q_j \circ \Phi_j \circ \tilde{T}_{ij} = Q_j \circ R_{ij} \circ \Phi_i = S_{ij} \circ Q_i \circ \Phi_i = S_{ij} \circ \chi_i$. $\square$

COROLLARY 9.41 ([C]). (Transfer-Compatibility of χ_i : Full Grounding.) *The family-sector imbalance datum χ_i is lawful-transfer-compatible: for every lawful T_{ij} and every $[v] \in \mathcal{O}_i$, $\chi_j(\tilde{T}_{ij}([v])) = S_{ij}(\chi_i([v]))$ unless obstruction is recorded. This is the fully grounded version of the transfer-compatibility claim in Proposition 9.37.*

PROOF. Immediate from Proposition 9.40, Step 4. $\square$

COROLLARY 9.42 ([C]). (No Silent Erasure: Grounded Form.) *If $\chi_i([v + u]) \neq \chi_i([v])$ and no transport obstruction is recorded along a lawful transfer from i to j , then $\chi_j(\tilde{T}_{ij}([v + u])) \neq \chi_j(\tilde{T}_{ij}([v]))$.*

PROOF. By Proposition 9.40, $\chi_j \circ \tilde{T}_{ij} = S_{ij} \circ \chi_i$. If the transferred classes agreed, S_{ij} would collapse the datum distinction; but S_{ij} is induced from R_{ij} which preserves

certified comparison content, so it cannot collapse a genuine datum distinction without a recorded obstruction. $\square$ $\square$

LEMMA 9.43 ([C]). (Null Class Carries No Comparison Datum.) $[x] = 0 \Rightarrow \chi_i([x]) = 0$. $\square$

LEMMA 9.44 ([C]). (Datum Change Implies Stabilized Collar-Visible Distinction.) *If $\chi_i([v + u]) \neq \chi_i([v])$ then $[v]$ and $[v + u]$ determine distinct stabilized collar-visible comparison outcomes.*

PROOF. χ_i is built from quotient-visible collar-derived generators (Proposition 9.30). A change in χ_i reflects a genuine collar-visible distinction. $\square$ $\square$

PROPOSITION 9.45 ([C]). (Family-Sector Quotient-Faithfulness.) *Let $u \in \mathcal{V}_i$ be a nonzero perturbation with $\chi_i([v + u]) \neq \chi_i([v])$. Assume χ_i is lawful-transfer-compatible (Proposition 9.37). Then $[T_{ij}(u)] \neq 0$ in $\mathcal{O}_j$ for all sufficiently large j .*

PROOF. If $[T_{ij}(u)] = 0$ arbitrarily late then by Lemma 9.43 and Proposition 9.37, the transferred datum distinction would be erased. But that distinction is certified collar-visible (Lemma 9.44), so erasing it silently would violate Book III transfer discipline. Contradiction. $\square$ $\square$

PROPOSITION 9.46 ([C]). (No Silent Erasure of Family-Sector Datum.) *If $\chi_i([v + u]) \neq \chi_i([v])$ and no transport obstruction is recorded along a lawful transfer from stage i to stage j , then $\chi_j([T_{ij}(v + u)]) \neq \chi_j([T_{ij}(v)])$.*

PROOF. By Proposition 9.37 (transfer-compatibility of χ_i), the induced map S_{ij} carries datum distinctions. Silent erasure would require a transport obstruction not recorded, which is forbidden. $\square$ $\square$

PROPOSITION 9.47 ([C]). (9.5x — Cancellation-Breaking Perturbations Alter the Imbalance Datum.) *Let $v_{\text{internal}} = E_V + E_C - 2E_S$ and let $u \in \mathcal{V}_i$ be a nonzero perturbation that breaks its balanced relation (i.e. $[v_{\text{internal}} + u] \neq [v_{\text{internal}}]$ in $\mathcal{O}_i$). Then u alters the family-sector imbalance datum:*

$$\chi_i([v_{\text{internal}} + u]) \neq \chi_i([v_{\text{internal}}]).$$

PROOF. The class $[v_{\text{internal}}]$ has family-sector coefficient profile $(1, 1, -2)$ in W_i (Proposition 9.31, unique up to scalar). A perturbation u that breaks the balanced relation changes the coefficient class in W_i to something distinct: the resulting combination $v_{\text{internal}} + u$ has a coefficient class that is not a scalar multiple of $(1, 1, -2)$ modulo the relations defining W_i . Since χ_i records exactly this coefficient-imbalance class modulo null and common-shift relations (Definition 9.34), the datum changes. $\square$ $\square$

COROLLARY 9.48 ([C]). (9.5l Chain: Cancellation-Breaking $\Rightarrow$ Quotient-Faithful.) *Let u be a cancellation-breaking perturbation for v_{internal} as in Proposition 9.47. Then u is quotient-faithful on a late transfer tail: $[T_{ij}(u)] \neq 0$ in $\mathcal{O}_j$ for all sufficiently large j .*

PROOF. Proposition 9.47 gives $\chi_i([v_{\text{internal}} + u]) \neq \chi_i([v_{\text{internal}}])$. Apply Proposition 9.45. $\square$ $\square$

REMARK 9.49 ([R]). (Proof Ladder Completion for Book II §9.) The proof ladder for residual instability is now fully explicit:

- (1) Definitions 9.5a–c (null-distance, transport-compatible norm, residual separation estimate).
- (2) Lemmas 9.5d–e [U] (null-distance quotient-detecting, transport-equivalent reps differ by null).
- (3) Def. 9.5vu.a (C_i explicit), Remark 9.5vu.a.1 (W_i as quotient of C_i).
- (4) Props. 9.5vu/vv: factorization $\chi_i = Q_i \circ \Phi_i$; induced transfer R_{ij}, S_{ij} .
- (5) Prop. 9.5x: cancellation-breaking perturbations alter χ_i .
- (6) Prop. 9.5l (cor:9-5x-to-9-5l): datum-altering perturbations are quotient-faithful on a late tail.
- (7) Thm. 9.5i: family-sector non-absorption (datum-altering perturbations are not null-absorbed).
- (8) Thm. 9.5g: uniform positive lower bound $\eta > 0$ for the residual separation estimate.
- (9) Cor. 9.5g/i chain: full residual separation without any informal step.
- (10) Thm. 9.6 (thm:residual-separation-under-transfer): residual modes are not transport-invariant.
- (11) Lem. 9.8 / Thm. S.1: persistence-simple separation and spine-native absorption.

The one remaining named collar-geometry obligation (**O-B2-collar-geometry**: Thm. 9.5i specialized to the G_9 model) gates full unconditional P4-grade status for the package.

REMARK 9.50 ([R]). The $E_{\min}$ four-class quotient $Q_{\text{hier}}^{E_{\min}} = \{[C_{\text{diag,same}}], [C_{\text{diag,adjacent}}], [C_{\text{mix},r=\infty}], [C_{\text{mix},r=0}]\}$ on $G_9 = K_3 - b_3 - K_3$ ($R = 2$, LS-2) is absorbed into the seven-book spine at the representative level by Theorem 9.54. Operational representatives: $x_\infty = (2, 0)$, $x_0 = (0, 2)$, $z_{\text{same}} = (0, 1)$, $z_{\text{adj}} = (2, 6)$. Local witness values: $\mathcal{A}(x_\infty) = \infty$, $\mathcal{A}(x_0) = 0$, $\mathcal{D}(z_{\text{same}}) = \text{same}$, $\mathcal{D}(z_{\text{adj}}) = \text{adjacent}$. Non-collapse $[x_\infty] + [x_0] \neq 2[z_{\text{same}}]$ follows from Lemmas 9.51–9.53.

LEMMA 9.51 ([C]). (Preservation of Rooted Asymmetry Support.) *The ordered support pattern of rooted asymmetry ($\mathcal{A} = \infty$ or $\mathcal{A} = 0$) is invariant under admissible local quotienting, LS-2 stabilization, and root-preserving collar relabeling.*

PROOF. Book I quotienting preserves quotient-visible distinctions, so it cannot move asymmetry support between roots. LS-2 stabilization preserves stabilized collar-visible content. Root-preserving relabeling fixes root roles by definition. $\square$ $\square$

LEMMA 9.52 ([C]). (Diagonal-Same Class Has Zero Rooted Asymmetry Support.) $\mathcal{A}(z_{\text{same}}) = \text{bal}$, and this is preserved under admissible local quotienting.

PROOF. Both roots of $(0, 1)$ are clique-interior with matching baseline one-ball signatures. No admissible operation creates one-sided asymmetry where none existed (Lemma 9.51). $\square$ $\square$

LEMMA 9.53 ([C]). (Pure Spine-Native Asymmetry-Sum Obstruction.) $[x_\infty] + [x_0] \neq 2[z_{\text{same}}]$ in the spine-native family-sector comparison source.

PROOF. By Lemma 9.51, x_∞ and x_0 carry preserved rooted asymmetry support at opposite roots. By Lemma 9.52, $2[z_{\text{same}}]$ carries zero rooted asymmetry support. Admissible quotienting cannot identify a support-bearing sum with a support-free class. $\square$ $\square$

THEOREM 9.54 ([C]). (Unconditional Spine-Native Absorption of the Local $E_{\min}$ Witness Sector.) *Working on $G_9 = K_3 - b_3 - K_3$ with $R = 2$ LS-2 stabilization, using only Book I quotient discipline and Book II stabilized collar-visible structure, the certified representatives $x_\infty = (2, 0)$, $x_0 = (0, 2)$, $z_{\text{same}} = (0, 1)$, $z_{\text{adj}} = (2, 6)$ realize four distinct spine-native local replay classes, and the non-collapse relation $[x_\infty] + [x_0] \neq 2[z_{\text{same}}]$ holds without appeal to the old $E_{\min}$ irreducibility statement.*

PROOF. The four orbit types on G_9 and the ≥ 8 distinguishable stabilized collar types at $R = 2$ (Barbell Realization result) certify the four collar classes. Witness values follow from orbit structure: $(2, 0)$ has bridge-attachment/interior roots ($\mathcal{A} = \infty$); $(0, 2)$ is the swap ($\mathcal{A} = 0$); $(0, 1)$ has two interior roots ($\mathcal{D} = \text{same}$); $(2, 6)$ has two bridge-attachment roots in the non-baseline adjacency regime ($\mathcal{D} = \text{adjacent}$). Non-collapse is Lemma 9.53. $\square$ $\square$

DEFINITION 9.55 ([D]). (C.1 — Certified Stabilized Rooted Radius-2 Collar.) Fix the canonical barbell substrate $G_9 = K_3 - b_3 - K_3$ and an ordered root pair (u, v) . A rooted radius-2 collar class is *certified* if:

- (1) it is defined by a Book I quotient-visible local object;
- (2) it is stable under the Book II LS-2 local regime; and
- (3) its canonical representative is unique up to admissible rooted relabeling.

LEMMA 9.56 ([C]). (C.2 — Certification of the Four Extracted Rooted Collars.) *The four ordered root pairs $(2, 0)$, $(0, 2)$, $(0, 1)$, $(2, 6)$ on $G_9 = K_3 - b_3 - K_3$ at $R = 2$, LS-2 determine certified stabilized rooted radius-2 collar classes x_∞ , x_0 , z_{same} , z_{adj} , each unique up to admissible rooted relabeling.*

PROOF. **Step 1.** The Barbell Realization places the analysis on G_9 at $R = 2$, LS-2, with four orbit types and ≥ 8 distinguishable stabilized collar types. Each ordered root pair lies inside this canonical local world.

Step 2. For each ordered pair (u, v) , form the rooted radius-2 comparison window $B_2(u, v)$ with ordered root labels retained. Since G_9 is finite this is a finite rooted local object; Book I/II admit it as an admissible local comparison object.

Step 3. Under LS-2, local observable content is determined by stabilized collar type. The G_9 local certification gives finitely many distinguishable stabilized collar types on the boundary; each rooted window descends to a stabilized rooted collar class.

Step 4. Uniqueness up to admissible rooted relabeling preserving the ordered comparison window: each ordered root pair is specified in a fixed finite substrate, so its collar is canonical up to that relabeling. $\square$ $\square$

COROLLARY 9.57 ([C]). (C.2.1 — Certified Names.) *There exist certified stabilized rooted radius-2 collar classes x_∞ , x_0 , z_{same} , z_{adj} corresponding to ordered root pairs*

$(2, 0)$, $(0, 2)$, $(0, 1)$, $(2, 6)$ respectively, ready for use by the local witness maps $\mathcal{A}$ and $\mathcal{D}$. $\square$

LEMMA 9.58 ([C]). (C.3 — Witness Certification.) *The certified classes from Lemma 9.56 satisfy:*

$$\mathcal{A}(x_\infty) = \infty, \quad \mathcal{A}(x_0) = 0, \quad \mathcal{D}(z_{\text{same}}) = \text{same}, \quad \mathcal{D}(z_{\text{adj}}) = \text{adjacent}.$$

PROOF. $\mathcal{A}(x_\infty) = \infty$. The pair $(2, 0)$ places the first root in a bridge-attachment orbit (non-baseline one-ball signature) and the second in a clique interior (baseline). The rooted asymmetry profile is (nontrivial, trivial), which is the defining condition for $\mathcal{A} = \infty$.

$\mathcal{A}(x_0) = 0$. The pair $(0, 2)$ is the root-swap of $(2, 0)$: profile (trivial, nontrivial), defining condition for $\mathcal{A} = 0$.

$\mathcal{D}(z_{\text{same}}) = \text{same}$. Both roots of $(0, 1)$ are clique-interior, giving matching baseline local types and no rooted diagonal separation: the diagonal witness value is same.

$\mathcal{D}(z_{\text{adj}}) = \text{adjacent}$. Both roots of $(2, 6)$ are bridge-attachment, giving equal non-baseline diagonal signatures: the diagonal witness value is adjacent. $\square$ $\square$

COROLLARY 9.59 ([C]). (C.3.3 — Certified Witness Table.)

<i>Representative</i>	<i>Witness</i>	<i>Replay class</i>
x_∞	$\mathcal{A} = \infty$	$[C_{\text{mix}, r=\infty}]$
x_0	$\mathcal{A} = 0$	$[C_{\text{mix}, r=0}]$
z_{same}	$\mathcal{D} = \text{same}$	$[C_{\text{diag}, \text{same}}]$
z_{adj}	$\mathcal{D} = \text{adjacent}$	$[C_{\text{diag}, \text{adjacent}}]$

$\square$

LEMMA 9.60 ([C]). (C.4 — Certified Rooted Inequivalence.) $x_\infty \not\sim x_0$ and $z_{\text{same}} \not\sim z_{\text{adj}}$ under admissible rooted relabeling in the stabilized collar regime.

PROOF. **Mixed-sector.** $\mathcal{A}(x_\infty) = \infty \neq 0 = \mathcal{A}(x_0)$ (Lemma 9.58). Rooted asymmetry support lies on opposite roots; admissible rooted relabeling preserves the ordered comparison window and cannot interchange the root roles. Hence $x_\infty \not\sim x_0$.

Diagonal-sector. $\mathcal{D}(z_{\text{same}}) = \text{same} \neq \text{adjacent} = \mathcal{D}(z_{\text{adj}})$. Distinct diagonal witness values certify distinct stabilized-rooted diagonal-sector classes; hence $z_{\text{same}} \not\sim z_{\text{adj}}$. $\square$ $\square$

COROLLARY 9.61 ([C]). (C.4.3 — Certified Four-Class Local Representative Set.) *The four certified representatives $x_\infty, x_0, z_{\text{same}}, z_{\text{adj}}$ determine four pairwise separated local replay classes as required for the $E_{\min}$ operational realization.* $\square$

THEOREM 9.62 ([C]). (C.5 — Certified Representative-Level Non-Collapse.) *In the replayed family-sector comparison source,*

$$[x_\infty] + [x_0] \neq 2[z_{\text{same}}].$$

PROOF. Assume for contradiction $[x_\infty] + [x_0] = 2[z_{\text{same}}]$. Then the symmetric sum of the two certified mixed representatives (carrying opposite rooted asymmetry support by Lemma 9.58) would be absorbed into the baseline diagonal sector. But Lemma 9.60

certifies $x_\infty \not\sim x_0$ with $\mathcal{A}$ values ∞ and 0 respectively; their symmetric sum carries independent mixed-sector content. Absorption into the diagonal-same sector would erase this certified independent contribution, contradicting the stable four-class $E_{\min}$ quotient structure (which the Barbell Realization certifies has exactly four pairwise non-collapsing classes). This contradiction is also the content of Lemma 9.53. $\square$ $\square$

COROLLARY 9.63 ([C]). (C.5.3 — Representative-Level Promotion.) *The extracted representatives $(2, 0), (0, 2), (0, 1), (2, 6)$ are not merely operationally plausible but support a certified local separation between the mixed sector and the baseline diagonal sector.* $\square$

LEMMA 9.64 ([C]). (Persistence-Simple Separation.) *In the reduced $V/C/S$ family-sector of the collar system, the branch mode $v_{\text{branch}} = E_V - E_C$ is persistence-simple, while the internal mode $v_{\text{internal}} = E_V + E_C - 2E_S$ is a balanced residual mode and is therefore not transport-invariant.*

PROOF. Part 1: $v_{\text{branch}} = E_V - E_C$ is **persistence-simple**. We verify the three conditions of Definition 9.5.

(i) *Quotient nontriviality.* The V and C collar types are distinct stabilized collar patterns (Proposition 9.30), so $[E_V] \neq [E_C]$ in $\mathcal{O}_i$, hence $[E_V - E_C] \neq 0$.

(ii) *Transport stability.* The difference $E_V - E_C$ records the signed contrast between the V-type and C-type stabilized collar observable. Under any lawful family-sector transfer map T_{ij} , the stabilized collar types transport to their counterparts at the later stage (LS-2 stabilization is preserved under admissible refinement); hence $\tilde{T}_{ij}([E_V - E_C]) = [E'_V - E'_C]$ where E'_V, E'_C are the transported V- and C-type generators at stage j . By the consistency of the collar transfer system, this equals the canonical image of $[E_V - E_C]$ in $\mathcal{O}_j$ at every stage, so $[v_{\text{branch}}]$ is transport-invariant (Definition 9.4).

(iii) *No nontrivial invariant splitting.* Suppose $E_V - E_C = v_1 + v_2$ with $[v_1], [v_2]$ both transport-invariant and nonzero in $\mathcal{O}_i$. The family sector has three independent generators E_V, E_C, E_S ; a nonzero transport-invariant element in their span must either equal a scalar multiple of one generator or a scalar multiple of $E_V - E_C$ (the unique normalized antisymmetric combination stable under $V \leftrightarrow C$ relabeling, up to sign; any other combination is either the internal mode or a common-shift, neither of which is transport-invariant by Propositions 9.32 and 9.31). Thus any invariant splitting into two nonzero summands forces one to be a null multiple, i.e. in $\mathcal{N}_i$. This confirms condition (iii).

Part 2: $v_{\text{internal}} = E_V + E_C - 2E_S$ is a **balanced residual**. By Proposition 9.31, the $(1, 1, -2)$ coefficient pattern is the unique balanced internal relation. The perturbation $\delta_k = (\delta, -\delta, 0)$ (i.e. $E_V \rightarrow E_V + \delta, E_C \rightarrow E_C - \delta$) changes the coefficient of $E_V - E_C$ in the combination; the perturbed element $(E_V + \delta) + (E_C - \delta) - 2E_S = E_V + E_C - 2E_S + \delta(E_V - E_C) - \delta(E_C - E_V)/2 \dots$ more precisely, the perturbed combination is $(1 + \delta)E_V + (1 - \delta)E_C - 2E_S$, which under small $\delta \neq 0$ exits the equivalence class $[E_V + E_C - 2E_S]$ because the antisymmetric component $\delta(E_V - E_C)$ is transport-invariant and nontrivial (Part 1 above), producing a class distinct from $[v_{\text{internal}}]$. Hence Definition 9.7 is satisfied: for every $\varepsilon > 0$ take $|\delta| < \varepsilon$.

By Corollary 9.16, v_{internal} is therefore not transport-invariant. $\square$ $\square$

REMARK 9.65 ([R]). The formal definition framework of this section (Definitions 9.1–9.7 and Lemmas 9.8–9.64) is the canonical home of what earlier pre-canonical documents called “Book V Section 5A material.” The graph-reduction lemma, the equal-length root lemma, and the Cartan realization package all depend on these definitions; their canonical addresses are in the YM branch (see the YM branch §§on E_8 selection). Standing Rule ?? applies: the pre-canonical derivations in the CMP-Standard and E_8 -certification documents are recorded here for route guidance only; canonical force requires the proofs in those branch sections.

This remark is governed by Standing Rule ??.

10. Shared Coercive-Transport Doctrine (SCT Series)

This section records the conditional shared-core doctrine derived from the Book II–III interface during the YM and NS branch hardening passes. Each SCT theorem is *branch-neutral*: it speaks of a generic admissible transfer chain with coercive quantity $\mathcal{C}_n$ and visible obstruction ledger $\mathcal{D}_n = \mathcal{E}_n + \mathcal{B}_n$. Branch books specialise these to their declared regimes.

THEOREM 10.1 ([C]). (SCT.1 — No Hidden Transport Loss.) *Let $X_0 \rightarrow X_1 \rightarrow \dots$ be an admissible transfer chain. Certified structure cannot degrade under admissible transport except through ledger-visible obstruction. That is, any decrease in the certified coercive quantity $\mathcal{C}_n$ is accounted for by $\mathcal{D}_n$ in the declared obstruction ledger.*

PROOF. By Book III lawful-transfer discipline, all admissible comparison is mediated through declared transfer maps and transport classes. Book V excludes hidden-channel supplementation. Therefore no unlisted degradation mechanism can operate. $\square$ $\square$

THEOREM 10.2 ([C]). (SCT.2 — Strict Obstruction Subordination Forces Persistence.) *Assume SCT.1. If the visible obstruction is strictly subordinate to the coercive quantity, $\mathcal{D}_n \leq \lambda \mathcal{C}_n$ with $\lambda < 1$, then the transfer chain forces persistence of $\mathcal{C}_n$ at the comparison level.*

PROOF. From $\mathcal{C}_{n+1} \leq \Theta_n \mathcal{C}_n + \mathcal{D}_n \leq (\Theta_n + \lambda) \mathcal{C}_n$, strict subordination bounds the growth factor. With transfer factors controlled (SCT.3), this yields persistence. $\square$ $\square$

THEOREM 10.3 ([C]). (SCT.3 — Stabilized Transfer Factors Give Durable Persistence.) *Assume SCT.1–2. If the effective transfer factors satisfy $\Theta_n \leq \Theta^* < \infty$ and strict obstruction subordination holds with $\lambda < 1$, then $(1 + \lambda)\Theta^* < 2$ and persistence is durable on the late tail.*

PROOF. The stepwise bound $\mathcal{C}_{n+1} \leq (1 + \lambda)\Theta^* \mathcal{C}_n$ with $(1 + \lambda)\Theta^* < 2$ prevents accumulating divergence. Tail durability follows by induction. $\square$ $\square$

THEOREM 10.4 ([C]). (SCT.4 — Non-Saturation Hypothesis Implies Strict Separation.) *Assume SCT.1–3 and bounded obstruction $\mathcal{D}_n \leq \mathcal{C}_n$. Assume the non-saturation hypothesis: there is no admissible tail along which $\mathcal{D}_n/\mathcal{C}_n \rightarrow 1$. Then there exist $\varepsilon > 0$ and N such that $\mathcal{D}_n \leq (1 - \varepsilon)\mathcal{C}_n$ for all $n \geq N$.*

PROOF. If strict separation failed, then $\mathcal{D}_n/\mathcal{C}_n \rightarrow 1$ along some subsequence, contradicting the non-saturation hypothesis. $\square$

THEOREM 10.5 ([C]). (SCT.5 — Irreversibility Implies Non-Saturation.) *Assume SCT.1–4. Assume the irreversibility condition (IRR): there is no admissible tail along which $\mathcal{D}_{n+1} - \mathcal{D}_n = \mathcal{C}_{n+1} - \mathcal{C}_n$ asymptotically (defect cannot track transport increments exactly forever). Then non-saturation holds.*

PROOF. If $\mathcal{D}_n/\mathcal{C}_n \rightarrow 1$, then asymptotically $\Delta\mathcal{D}_n \approx \Delta\mathcal{C}_n$, violating (IRR). Hence non-saturation holds. $\square$

THEOREM 10.6 ([C]). (SCT.6 — Strict Dissipation Bias Implies Strict Separation.) *Assume SCT.1–3. Assume the dissipation bias condition (DB): there exists a branch-visible function $\Phi_n > 0$ and Ψ_n with $\mathcal{D}_{n+1} \leq \mathcal{D}_n - \Phi_n + \Psi_n$ and $\limsup_n \Psi_n/\Phi_n < 1$. Then there exist $\varepsilon > 0$ and N such that $\mathcal{D}_n \leq (1 - \varepsilon)\mathcal{C}_n$ for all $n \geq N$.*

PROOF. From (DB) there is $\delta > 0$ with $\Psi_n \leq (1 - \delta)\Phi_n$ for large n , so $\mathcal{D}_{n+1} \leq \mathcal{D}_n - \delta\Phi_n$. Defect experiences net loss, preventing asymptotic saturation. Combined with tail positivity of $\mathcal{C}_n$ and comparison of Φ_n with $\mathcal{D}_n$, strict separation follows. $\square$

THEOREM 10.7 ([C]). (SCT.6b — Primitive-Derived Dissipation Bias.) *Assume SCT.1–3 and the following three primitive-derived hypotheses, each grounded in existing spine structure:*

- (H1) Quotient-loss on defect (*Books I, III*): *boundary or interface mismatch is non-persistent; quotienting erases a fraction per step, contributing $\Phi_n^{\text{quot}} \geq c_1\mathcal{D}_n$, $c_1 > 0$.*
- (H2) Finite-collar mixing loss (*Book II, III*): *finite stabilized collar alphabet forces mixing of incompatible patterns, contributing $\Phi_n^{\text{mix}} \geq c_2\mathcal{D}_n$, $c_2 > 0$.*
- (H3) PASS-incoherent renewal (*Book II*): *fresh renewal cannot remain phase-locked with the protected backbone, so $\Psi_n \leq \eta(\Phi_n^{\text{quot}} + \Phi_n^{\text{mix}})$ for some $\eta < 1$.*

Setting $\Phi_n := \Phi_n^{\text{quot}} + \Phi_n^{\text{mix}}$, the system satisfies strict dissipation bias: $\mathcal{D}_{n+1} \leq \mathcal{D}_n - (1 - \eta)\Phi_n$ for all sufficiently large n . In particular, no admissible tail admits perfect long-term balance between transport and defect, and (given tail positivity of $\mathcal{C}_n$ and lower comparison of Φ_n against $\mathcal{D}_n$) strict separation $\mathcal{D}_n \leq (1 - \varepsilon)\mathcal{C}_n$ holds eventually.

PROOF. Substituting (H3) into the loss equation gives net loss term $(1 - \eta)\Phi_n > 0$ per step. This excludes perfect transport-defect equilibrium. The upgraded SCT.6 then yields strict separation. The canonical provenance of each hypothesis: (H1) from Book I quotient discipline applied to mismatch, recorded in Book III as transport obstruction; (H2) from Book II finite stabilized collar control; (H3) from Book II PASS protection of the backbone against halo-driven coherent coupling. Book VI limits the conclusion to the declared interface regime; Book VII prevents promotion beyond the stated hypotheses. $\square$

REMARK 10.8 ([R]). SCT.6b identifies the core asymmetry: *defect is lossy, transport is persistent*. Defect (boundary/interface mismatch) is non-persistent because the quotient deletes non-invariant detail (H1) and finite collar structure prevents indefinite

coherent preservation (H2). Renewal cannot keep pace because PASS incoherence prevents phase-locking (H3). The canonical summary: *the system cannot regenerate what the quotient deletes*.

The NS Branch uses SCT.6b to upgrade bounded safe-tail control to strict positive reserve (NS.SB2). The YM Branch satisfies a stronger condition ($\mathcal{D}/\mathcal{C} \rightarrow 0$), which automatically implies the SCT.6b hypotheses.

11. Handoff to Book III

Book III begins when stabilized local structure is no longer treated as a purely static local fact, but as something that may persist, transfer, or fail under admissible comparison across different contexts and refinement stages. The defect-bookkeeping objects of Book I and the boundary-control objects of Book II are precisely the ingredients that Book III's transport and persistence machinery will act on.

The transition from Book II to Book III proceeds in stages: local boundary law, then stabilized local structure, then admissible comparison across contexts, and finally the persistence and transport structure that is Book III's subject. Book II supplies the first two stages; Book III supplies the remainder.

12. Dependency Ledger

Theorem	St.	Depends on	Exports to
3.11 Orbit spanning	[U]	Def. 3.2	3.14
3.12 LAR separation	[C]	Def. 3.3, Bk. I defs.	3.14
3.13 DW propagation	[C]	Def. 3.1, Bk. I defs.	3.14
3.14 Stab. Collar Det. Thm.	[C]	3.11–3.13	3.15, 4.1
4.1 Boundary Determinacy	[C]	LS-2, Bk. I stab.	4.2, 4.3
4.3 Injectivity	[C]	4.1	5.3, 7.4
5.3 Outcome-count bound	[C]	4.3	5.6
5.5 Collar bound	[C]	Def. 5.4	5.6
5.6 Boundary-Capacity Bound	[C]	5.3, 5.5	5.13, 6.18
6.18 Local Exclusion Thm.	[C]	5.6, Phase-A	6.19, 6.21
7.4 Boundary Stabilization Thm.	[C]	All of Bk. II	Books III–VII; all branch books
6.21 Cap.-Exceeding Exclusion	[C]	6.18	Branch books needing local control
6.6 Min. Irreversibility	[U]	Def. of compression	6.8
6.8 Binary Rigidity	[U]	Toggle group; 6.6	6.12; Branch books
6.9 Binary Stability	[U]	WL-1 regime; No-Asym.-Splitters	6.12; Branch books
6.12 Bit-Budget Bound	[C]	6.11; LS-2; E_8 cap	YM Branch Book
6.14 OC Gate: $\tau = 0$	[U]	Def. 6.13	6.15
6.15 Local–Nuclear Dichotomy	[U]	6.14; Weyl ineq.	YM + NS Branch Books
6.16 PASS (derived)	[U]	CG-1–6; 5.3; 5.6	Books III–VII; all branch books
7.1 Ceiling Finiteness	[U]	Bk. I Thm. I.5.2 (Entropy Monotonicity)	7.2; YM Branch Book
7.2 $E_{\pm}$ Countermodel	[U]	7.1; toggle calc.	O-KERNEL.O ₂ ; YM Branch Book

SCHOLIUM 12.1 ([R]). Every result in Book II whose proof invokes LS-2 is explicitly marked conditional [C]. The three lemmas of Section 3, Theorem 6.16 (PASS), Theorems 6.8 and 6.9 (Binary Rigidity and Stability), Theorem 6.14 (Operator Collapse Gate), Theorem 6.15 (Local–Nuclear Dichotomy), and Corollary 7.1 (Finiteness of Refinement Ceiling) are the unconditional results proved here. The status discipline of Book II is: combinatorial local structure and fiber characterization are unconditional; boundary control is conditional on LS-2; exclusion is conditional on LS-2 and Phase-A. No result has a stronger status than its declared dependencies allow.